Unit 1

Digital Image Fundamentals

Outline

- Introduction
- Fundamental Steps in Digital Image Processing
- Components of an Image Processing System
- Image Sampling and Quantization
- Some Basic Relationships between Pixels
- connected component analysis

Introduction:

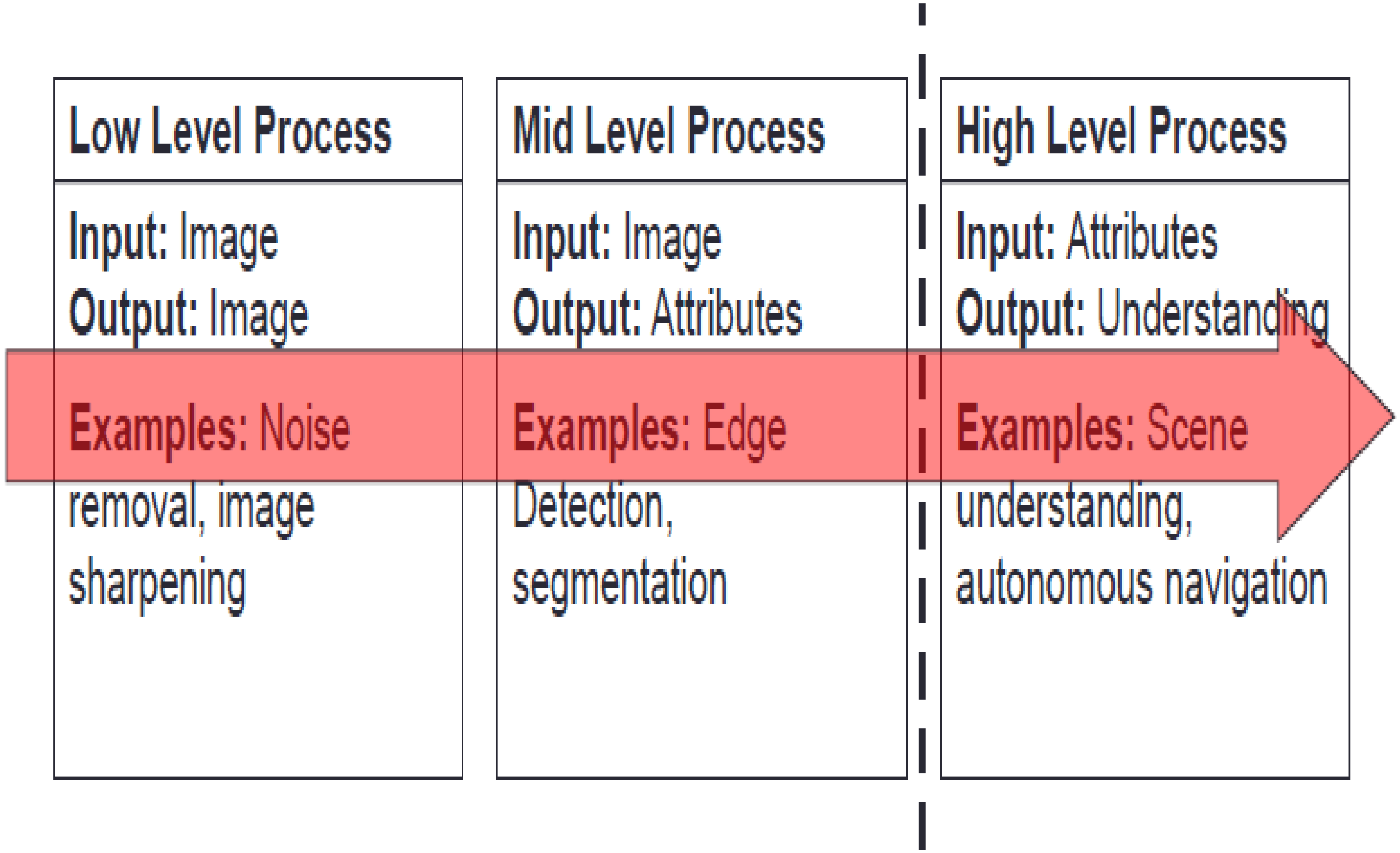
- Image is 2D function *f (x, y)*
  - *x* and *y* : Spatial co-ordinates, f : gray level
  - If *x, y,* and *f* are finite and discrete : Digital Image
- Each element in a digital image : Picture element or Image element or Pixel or pel
- Relationships among different fields

	Image	Model
Image	Image Processing	Computer Vision
Model	Computer Graphics	Computing Geometry

- **Computer Vision:**
  - Computers or machines are made to gain high-level understanding from the input digital images or videos
  - Purpose : automating tasks that the human visual system can do
  - It uses many techniques and Image Processing is just one of them

- Image Processing
  - Field of processing the raw input images to enhance them or preparing them to do other tasks
  - Image Processing is the subset of Computer Vision

- The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes.



- DIP: human is the final explainer

- Computer vision: computer is the final explainer

- Computer vision system needs the support of DIP

- DIP to computer vision is a process from low level to high level processing

Fundamental Steps in DIP

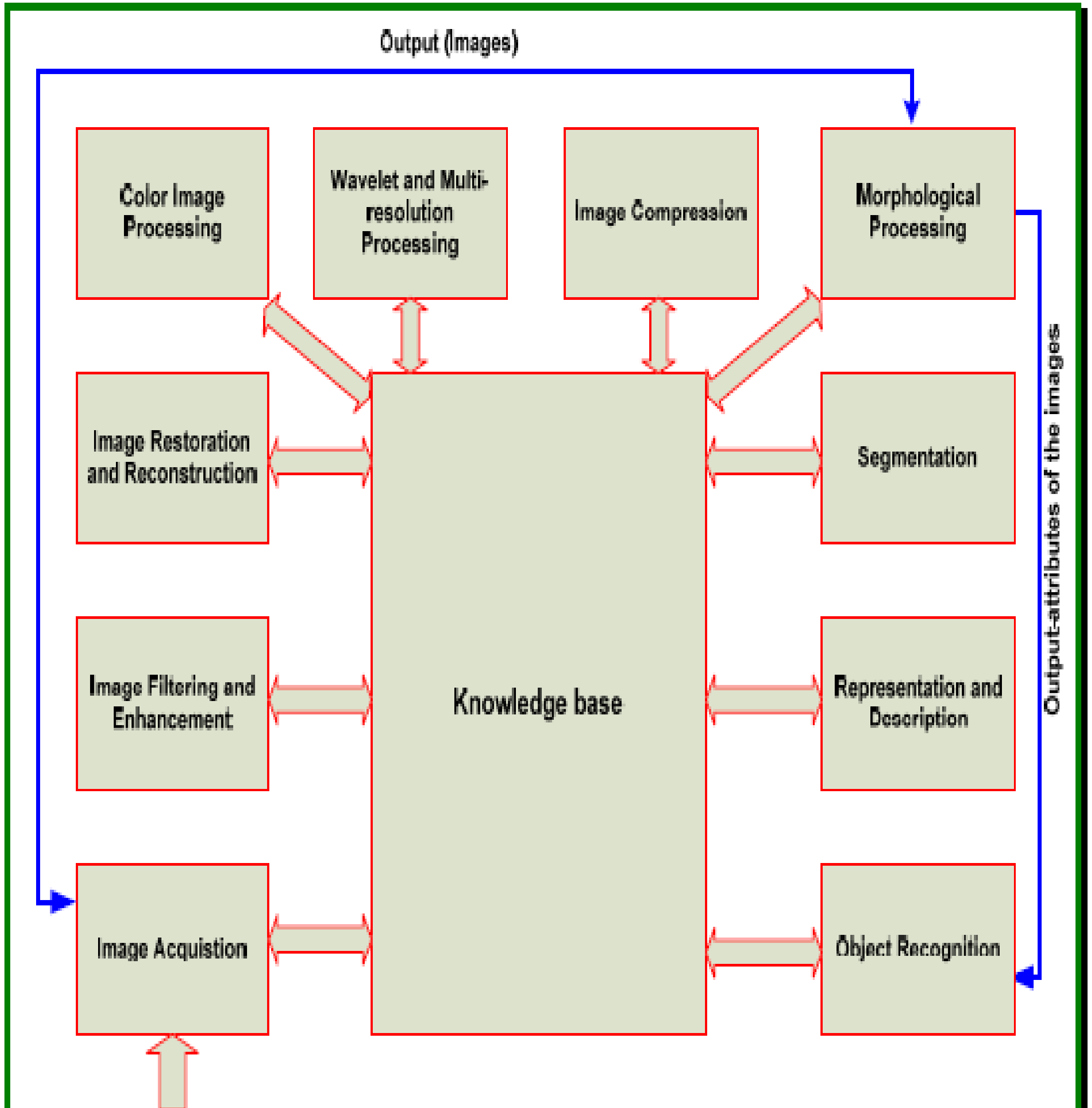
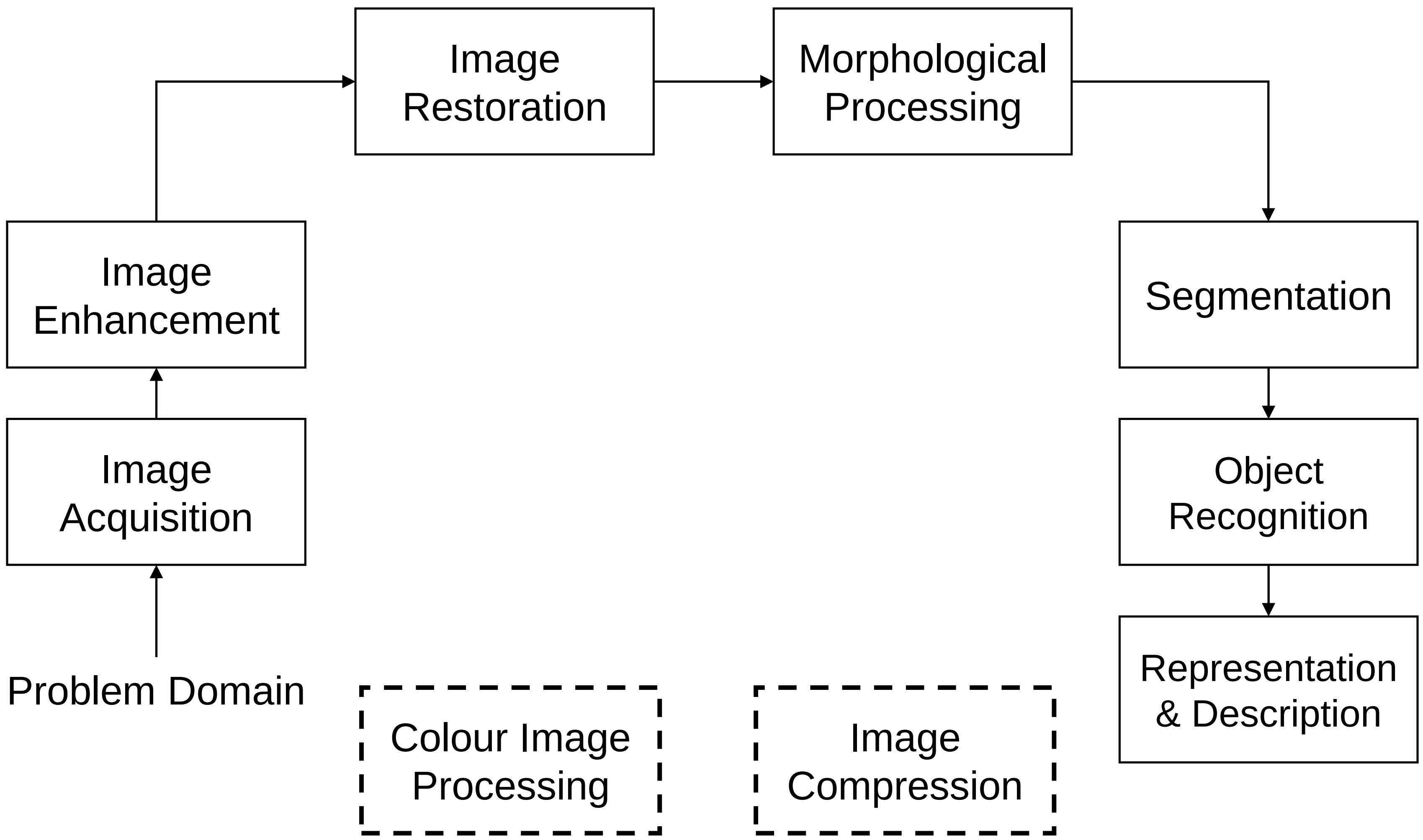
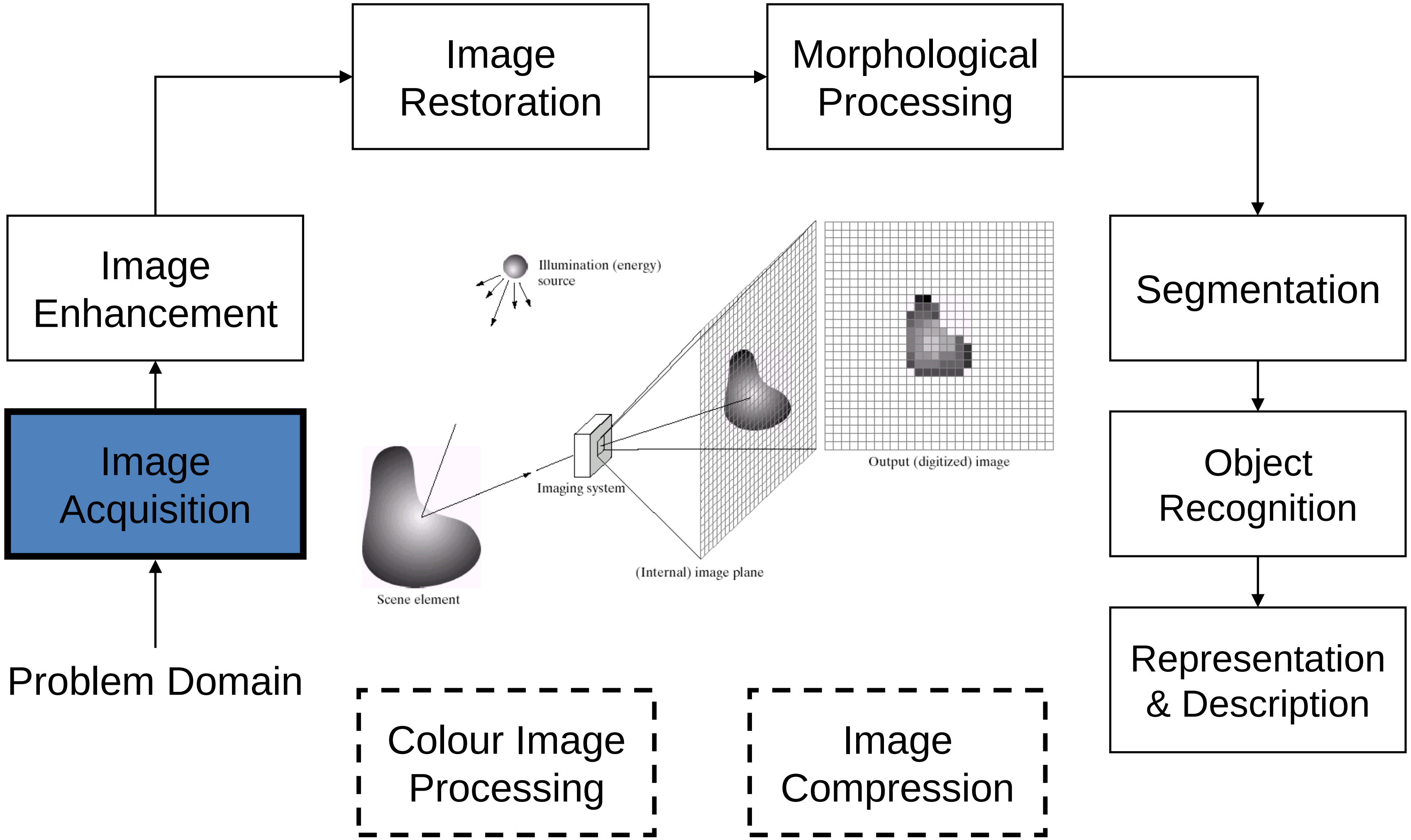


Figure 1-2

Key Stages in Digital Image Processing



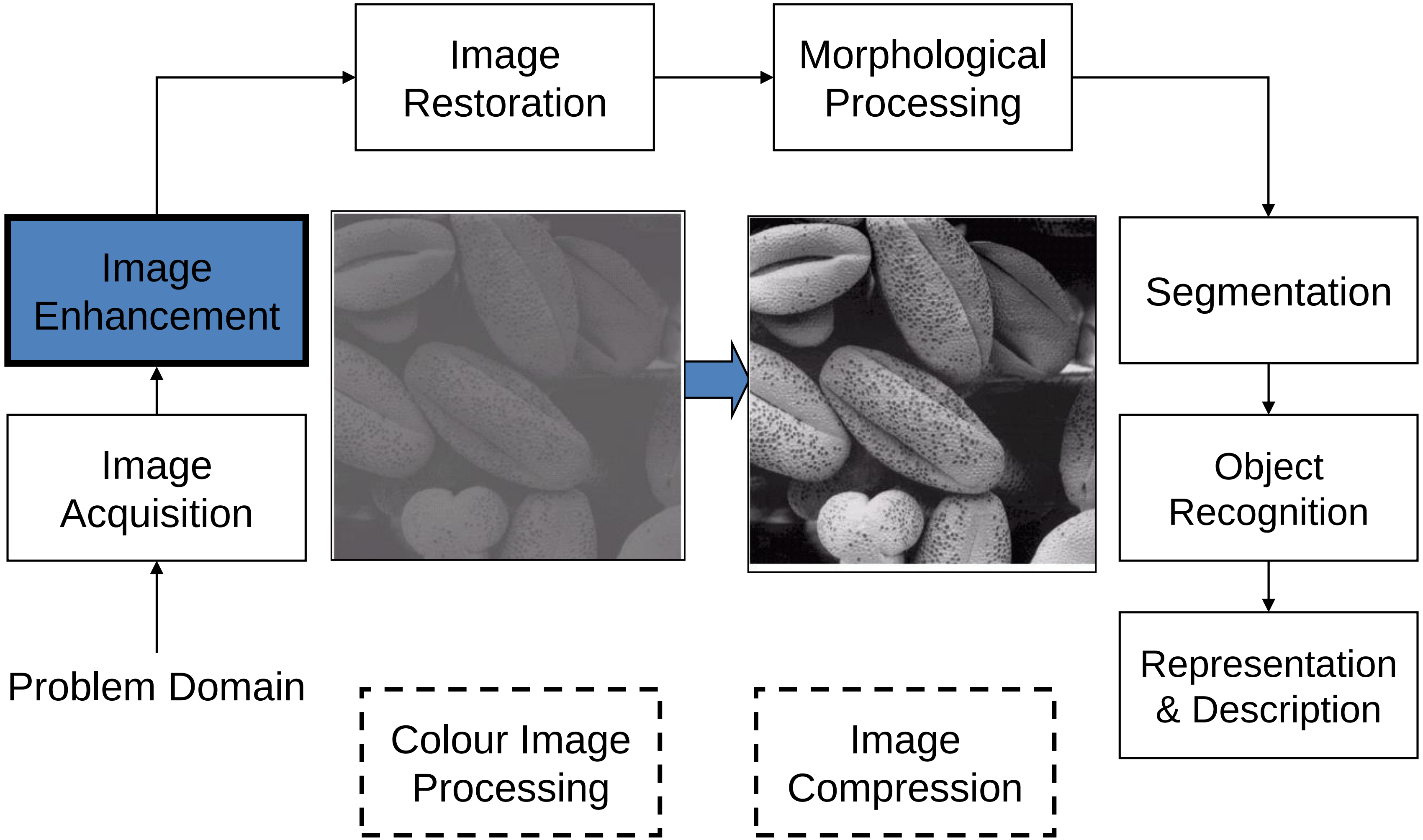
Key Stages in Digital Image Processing: Image Aquisition



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

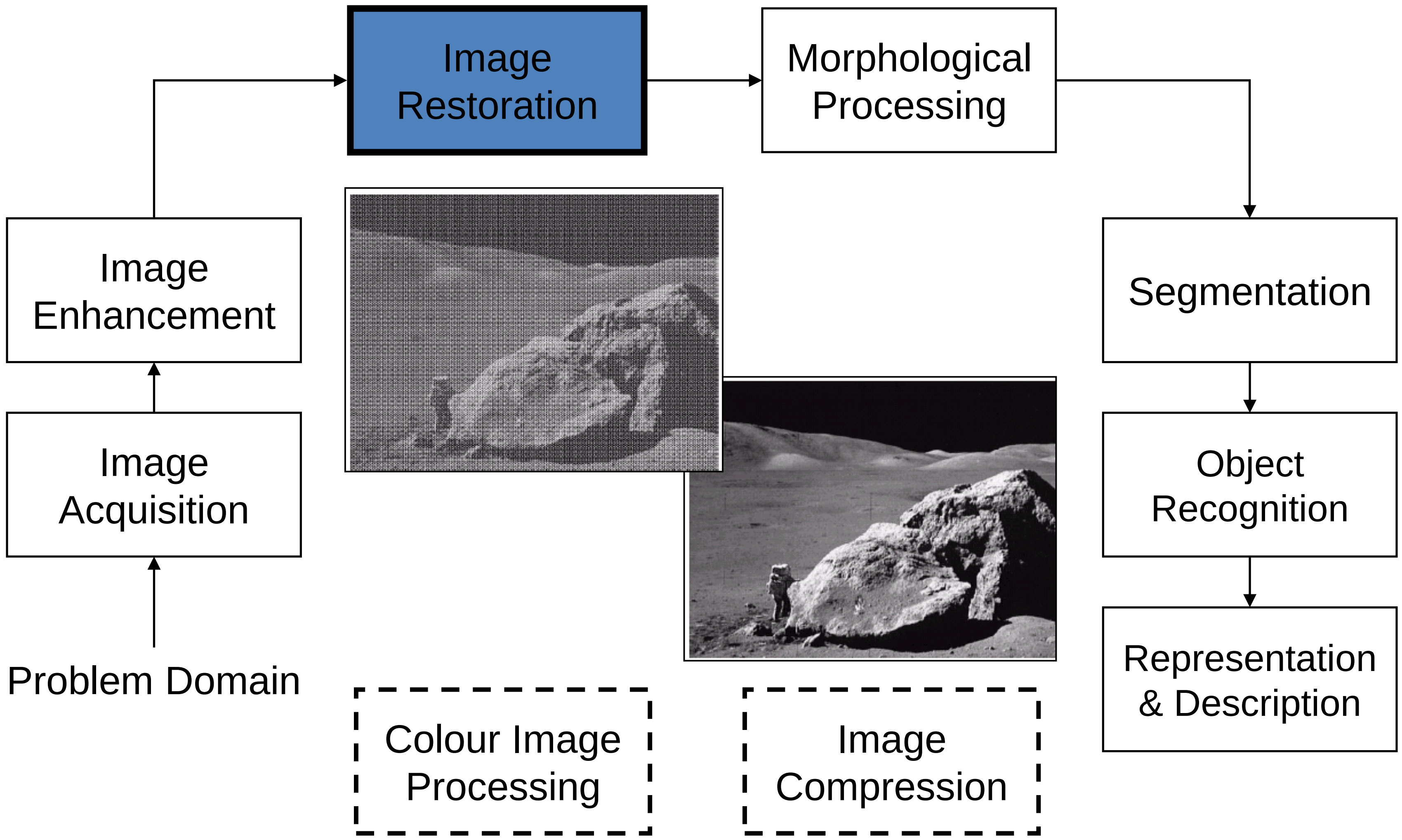
Key Stages in Digital Image Processing: Image Enhancement

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Key Stages in Digital Image Processing: Image Restoration

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Key Stages in Digital Image Processing: Morphological Processing

Image Acquisition

Image Enhancement

Image Restoration

Morphological Processing

Segmentation

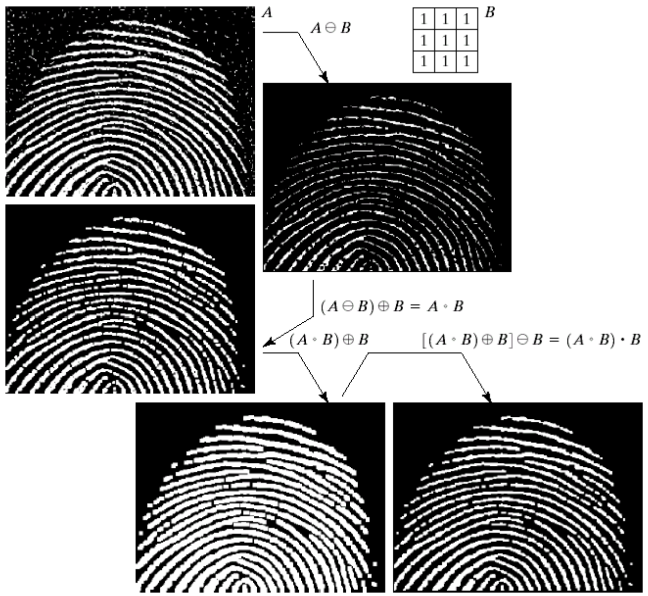
Object Recognition

Representation & Description

Problem Domain

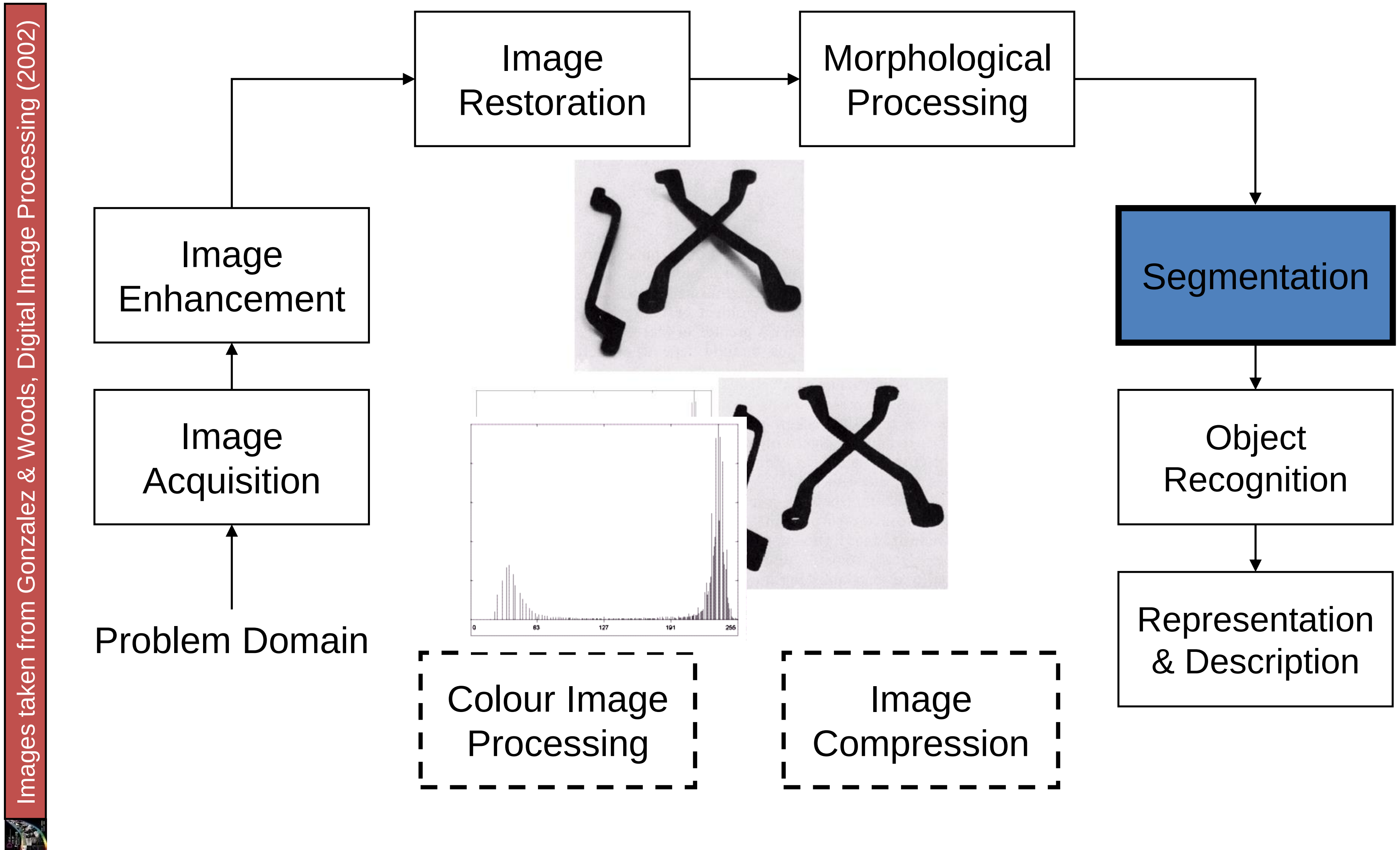
Colour Image Processing

Image Compression



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Key Stages in Digital Image Processing: Segmentation



Key Stages in Digital Image Processing: Object Recognition

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Restoration

Morphological Processing

Image Enhancement

Segmentation

Image Acquisition

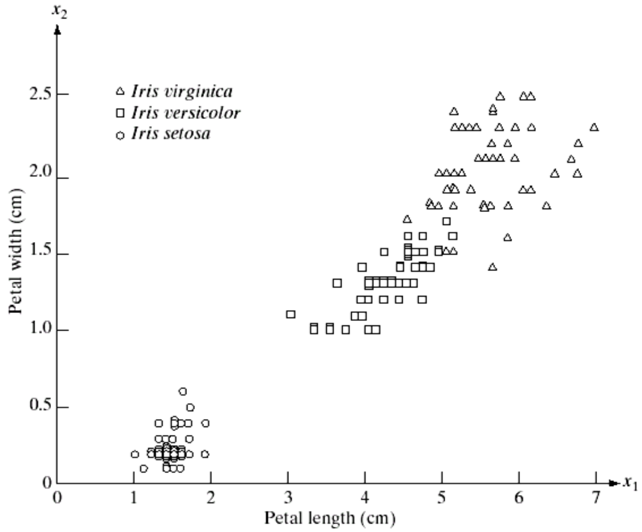
Object Recognition

Representation & Description

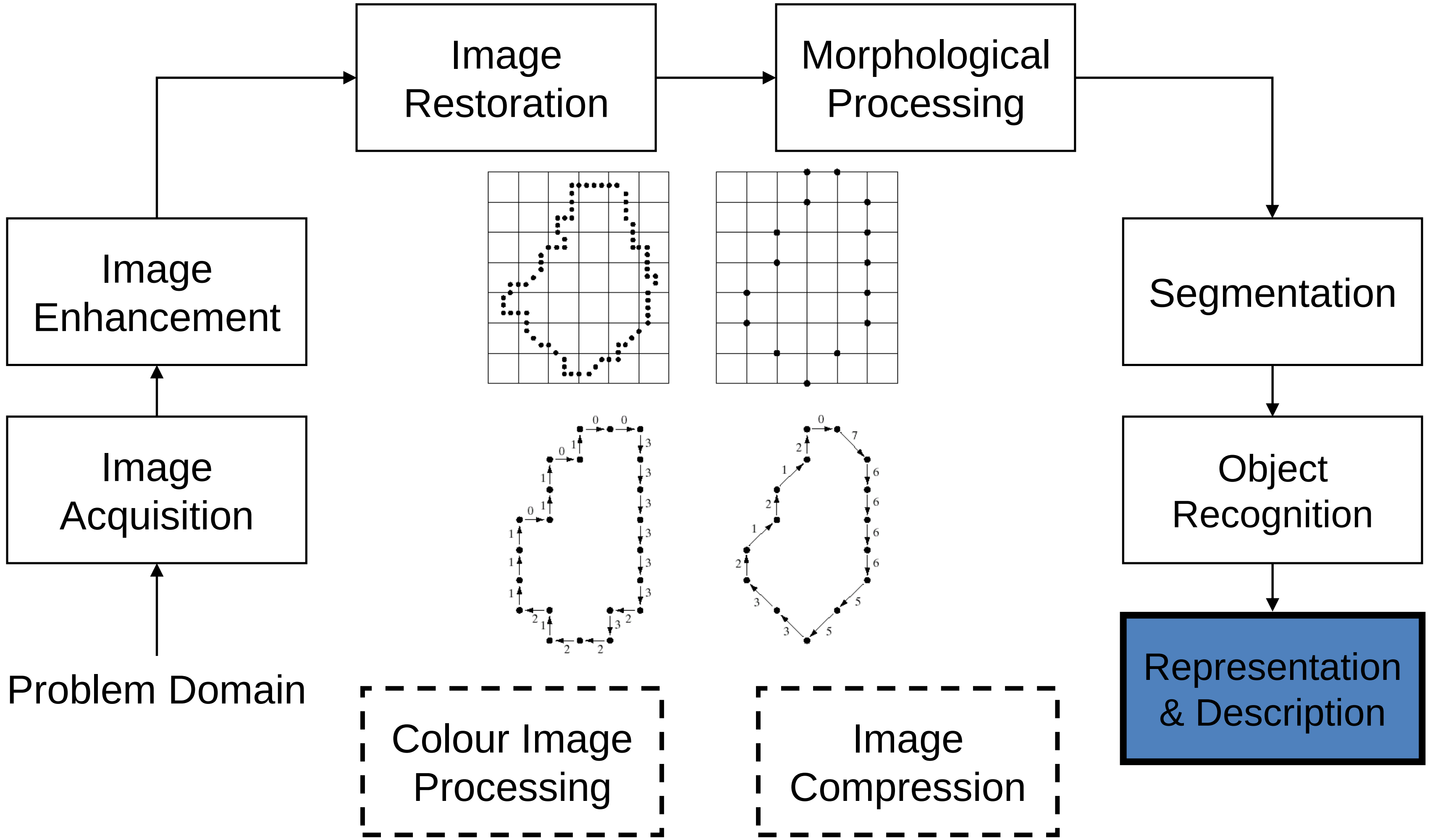
Problem Domain

Colour Image Processing

Image Compression

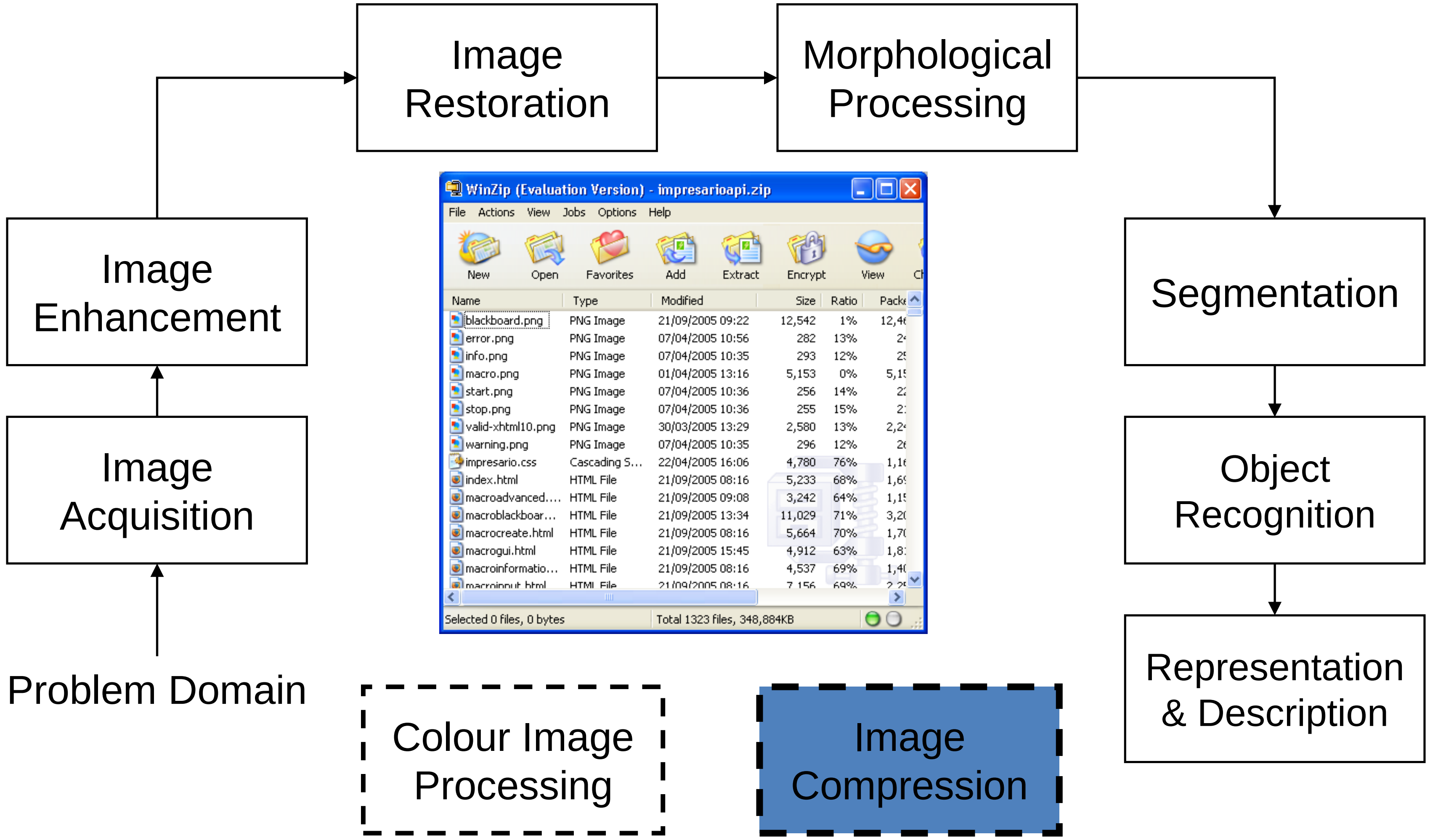


Key Stages in Digital Image Processing: Representation & Description



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Key Stages in Digital Image Processing: Image Compression



Key Stages in Digital Image Processing: Colour Image Processing

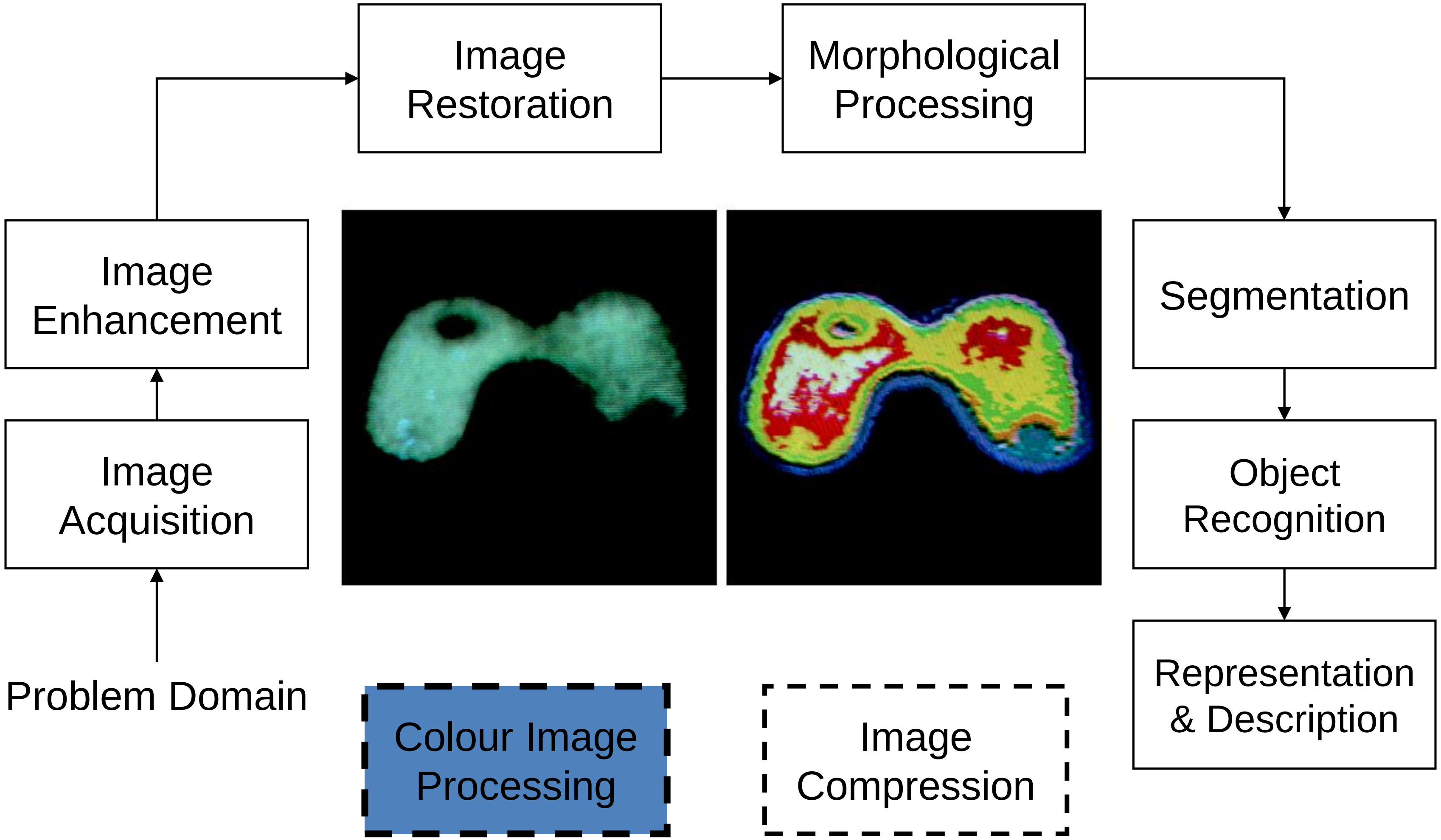


Image Enhancement

The principal objective of enhancement : Process an image so that the result is more suitable than the original image for a specific application

Image Restoration

Image restoration attempts to restore images that have been degraded

- Identify the degradation and try to reverse it
- Similar to image enhancement, but more objective

Morphological processing

Deals with tools for extracting image components that are useful in the representation and description of shape

Segmentation

Procedures which partition an image into its constituent parts or objects

Autonomous segmentation is one of the most difficult tasks rugged segmentation procedure brings the process towards successful solution

weak or erratic segmentation algorithms almost always guarantee eventual failure

Representation and description

- Always follows the output of a segmentation stage
- Segmentation o/p is raw pixel data, constituting either the boundary of a region or all the points in the region itself
- Boundary representation is appropriate when the focus is on external shape characteristics
- Regional representation is appropriate when the focus is on internal properties

Recognition

Process that assigns a label (e.g., “vehicle”) to an object based on its descriptors

Components of Image Processing System

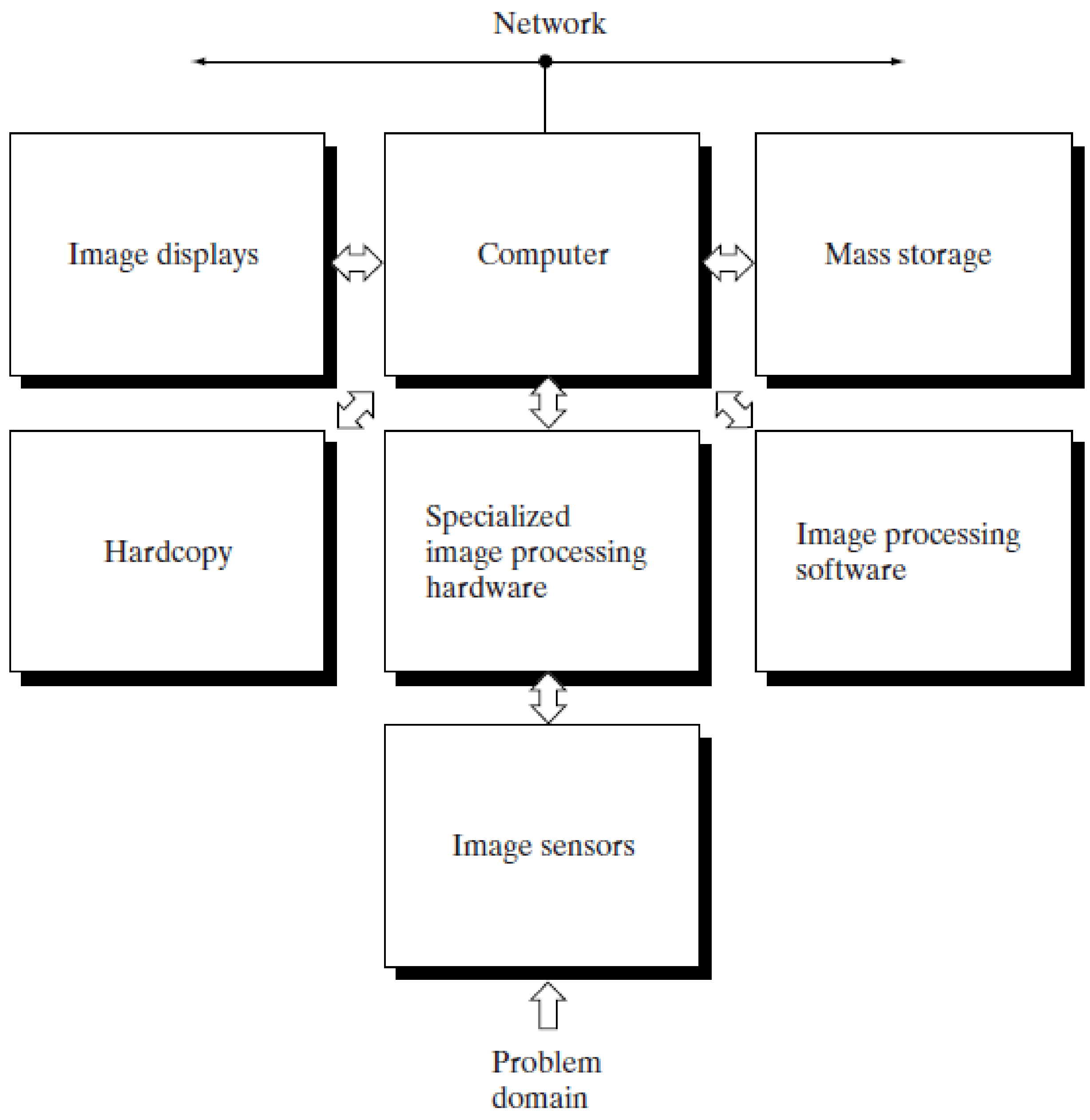
- Mid-1980’s image processing systems sold were substantial peripheral devices attached to host computers

- Late in 1980 and early 1990s, image processing hardware was in the form of single boards

- Large IP systems are developed for massive applications (processing satellite images)

- Trend is continued to miniaturize GP small computers with specialized image processing hardware

Components of Image Processing System contd..



Components of Image Processing System contd..

Sensing

- Two elements are required to acquire digital images
  - Physical device that is sensitive to energy radiated by the object
  - Digitizer, which converts output of physical device into digital form

Specialized image processing hardware

- Contains digitizer + hardware that performs primitive operations
  - e.g.: ALU is used for averaging images to reduce noise

Components of Image Processing System contd..

Computer

- It is an image processing system
- It is a GP computer ranging from PC to a supercomputer
- In dedicated applications, specially designed computers are used to achieve better performance

Software

- Software for IP consists of specialized modules that perform specific tasks

Components of Image Processing System contd..

Mass storage

Digital storage for IP applications fall into 3 principal categories:

- short term storage for use during processing
- on-line storage for relatively fast recall
- archival storage, characterized by infrequent access

Components of Image Processing System contd..

Short term storage is provided by

- Computer memory
- Specialized boards called frame buffers which give access at video rate

Online storage generally takes the form of magnetic disks or optical-media storage

- Characterized as frequent access

Archival storage is characterized by massive storage requirements

- But infrequent need for access

Components of Image Processing System contd..

Image displays

- Color TV monitors are used nowadays
- Monitors are driven by the outputs of image and graphics display cards
  - These cards are an integral part of the computer system

Components of Image Processing System contd..

Hard copy

- Devices for recording images
  - Include laser printers, film cameras, heat-sensitive devices, inkjet units, and digital units, such as optical and CD-ROM disks
- Film provides the highest possible resolution
  - However, paper is the obvious medium of choice for written material
- For presentations, images are displayed
  - on film transparencies
  - or in a digital medium if image projection equipment is used

Components of Image Processing System contd..

Networking

- Almost a default function in any computer system in use today
- Key consideration in image transmission is bandwidth
- In dedicated networks, this typically is not a problem
- Communications with remote sites via the Internet are not always efficient

A Simple Image Formation Model

- Images are denoted by two-dimensional functions of the form f(x, y)

- When an image is generated from a physical process, its values are proportional to energy radiated by a physical source

- f(x, y) must be nonzero and finite

$$0 < f(x, y) < \infty$$

A Simple Image Formation Model contd..

- The function f(x, y) may be characterized by two components:
  - The amount of source illumination incident on the scene being viewed - illumination component i(x, y)
  - The amount of illumination reflected by the objects in the scene - reflectance component r(x, y)
- The two functions combine as a product to form f(x, y)

$$f(x, y)=i(x, y)\ r(x, y)$$

where

$$0 < i(x, y) < \infty$$

$$0 < r(x, y) < 1$$

A Simple Image Formation Model contd..

- Reflectance is bounded by *0* (total absorption) and *1* (total reflectance)

- Intensity of a monochrome image at any coordinates (x_0, y_0) the gray level (ℓ) of the image at that point

$$\ell = f(x_0, y_0)$$

ℓ lies in the range

$$L_{min} \leq \ell \leq L_{max}$$

L_{min} must be positive

L_{max} must be finite

A Simple Image Formation Model contd..

$$L_{min}=i_{min}\, r_{min}$$

$$L_{max}=i_{max}\, r_{max}$$

- The interval $[L_{min}, L_{max}]$ is called the gray scale

- Common practice is to shift this interval numerically to the interval [0, L-1]

  $\ell = 0 \rightarrow$ black on gray scale

  $\ell = L-1 \rightarrow$ white on gray scale

- All intermediate values are shades of gray varying from black to white

Image Sampling and Quantization

- Acquisition of image → Sensors (camera)
- Output of most sensors is a continuous voltage waveform
- To create a digital image, convert the continuous sensed data into digital form
- Involves two processes:
  - **Sampling:** Digitizing the coordinate values
  - **Quantization:** Digitizing the amplitude values

Image Sampling and Quantization

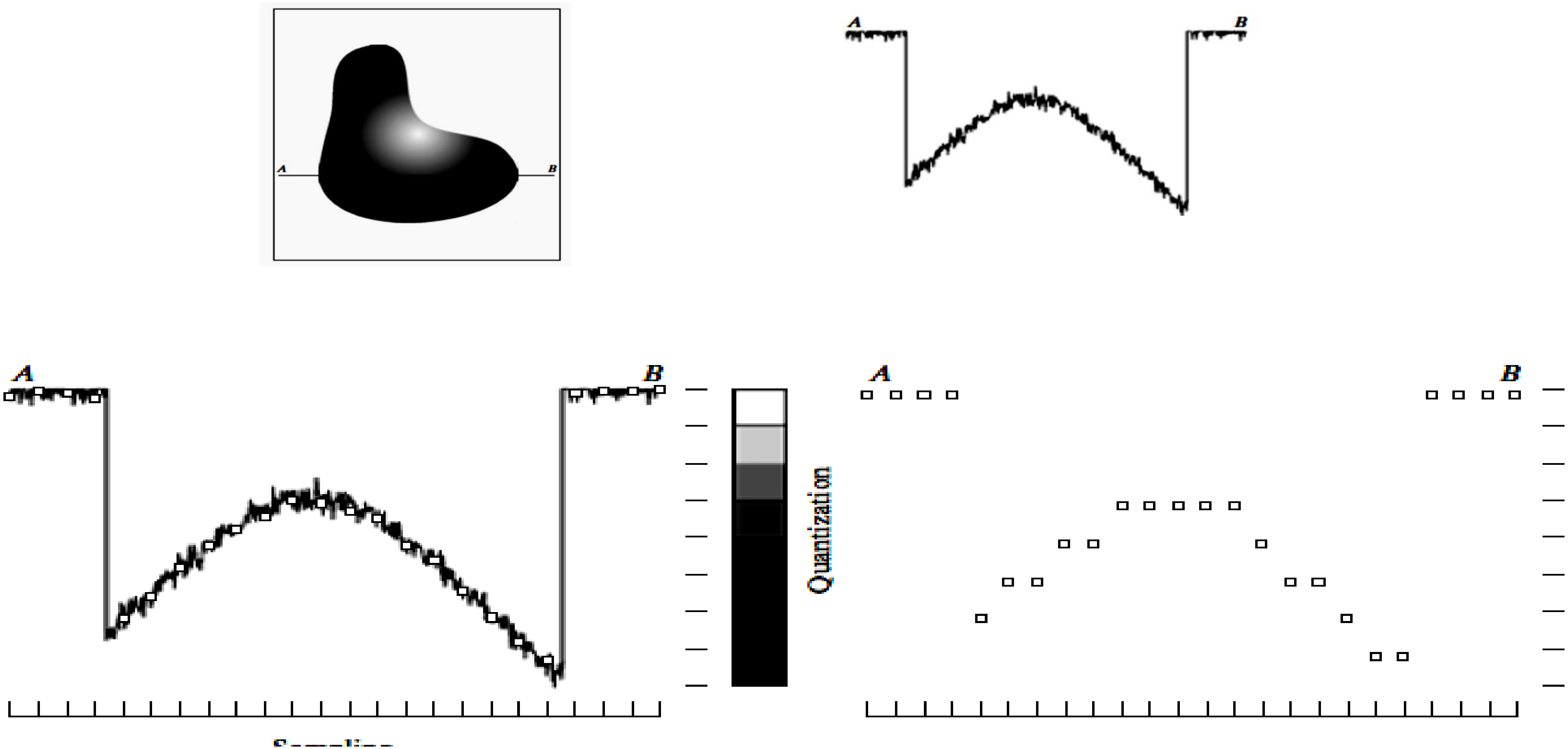
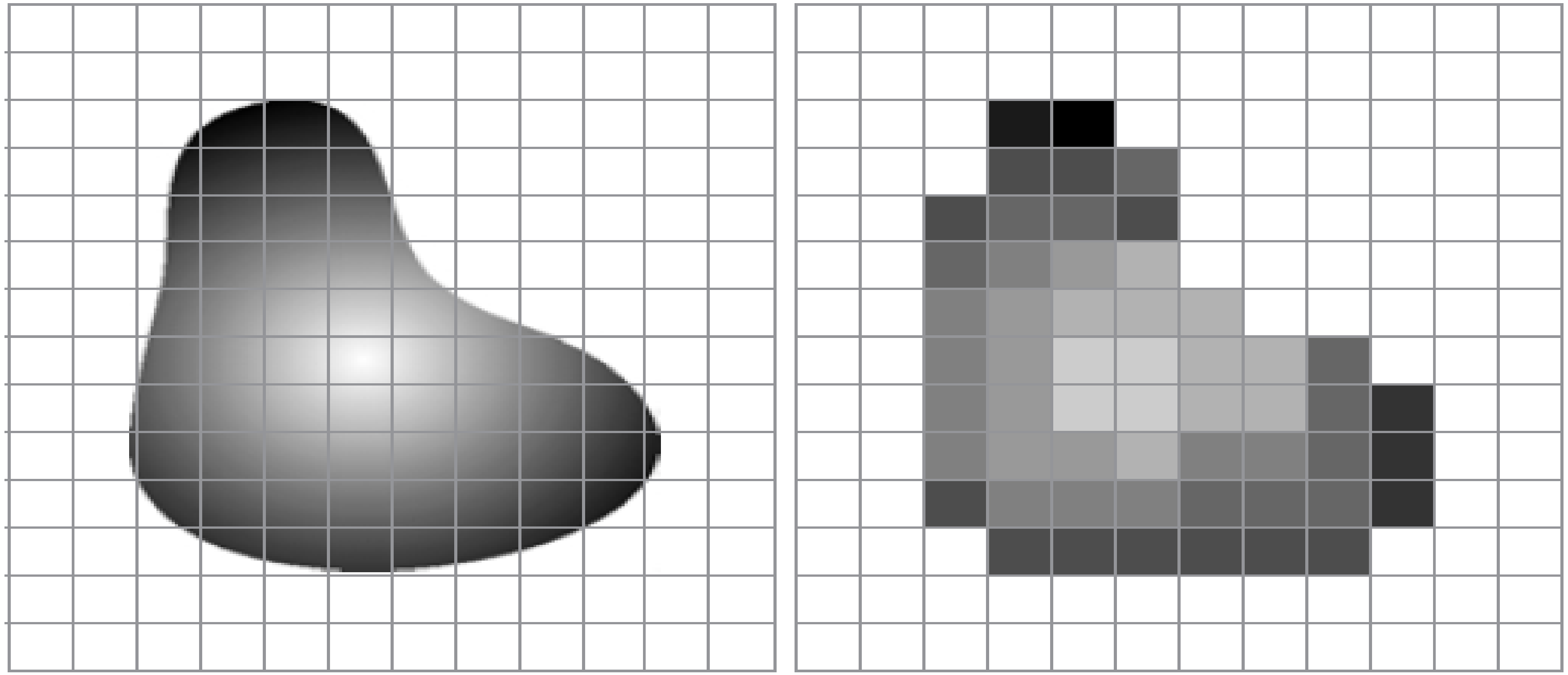


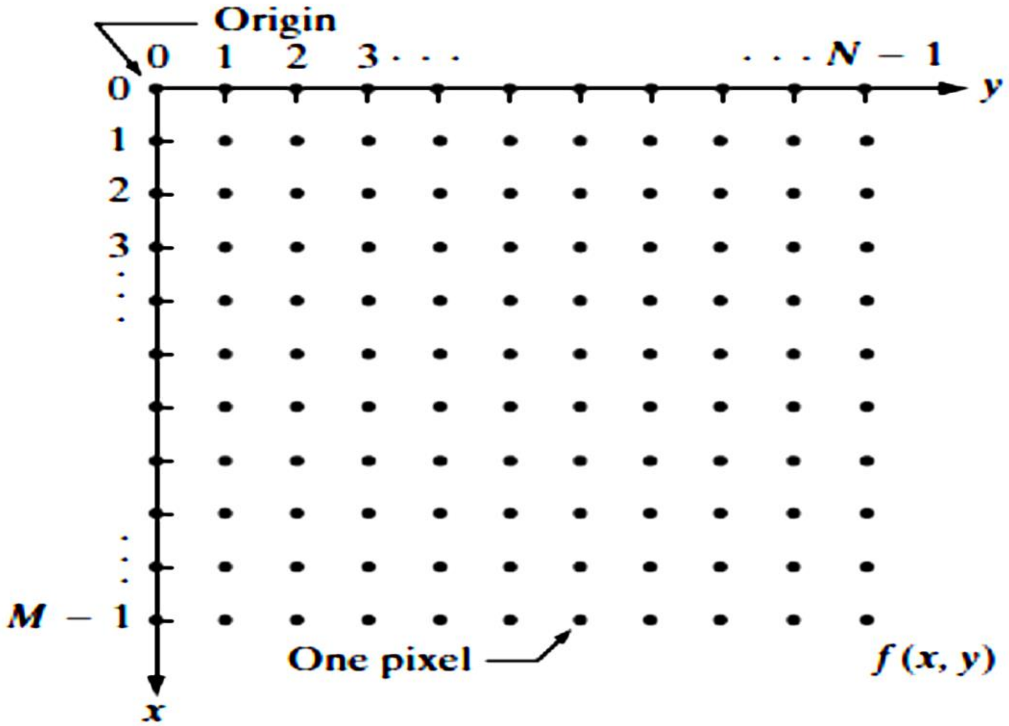
Image Sampling and Quantization contd..



(a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization

Representing Digital Images

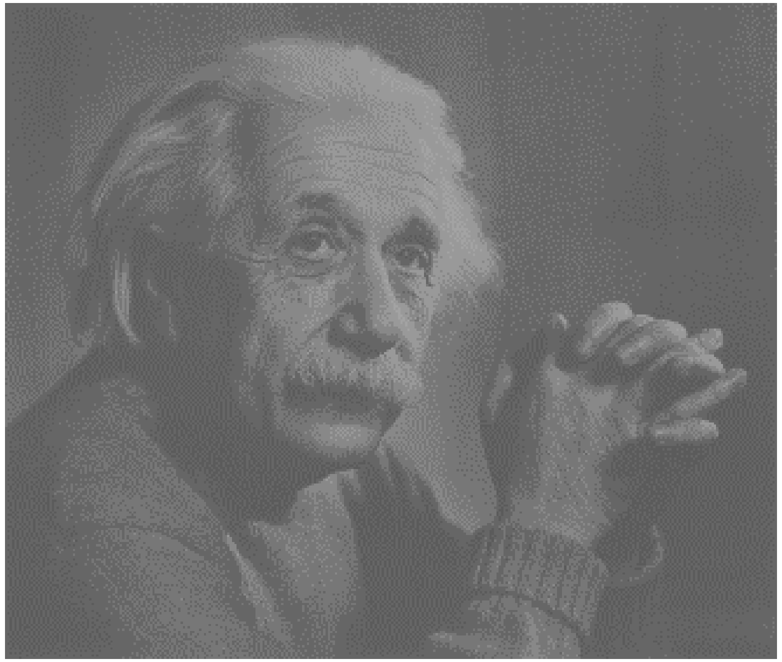
- Sampling and quantization results in a matrix of real numbers



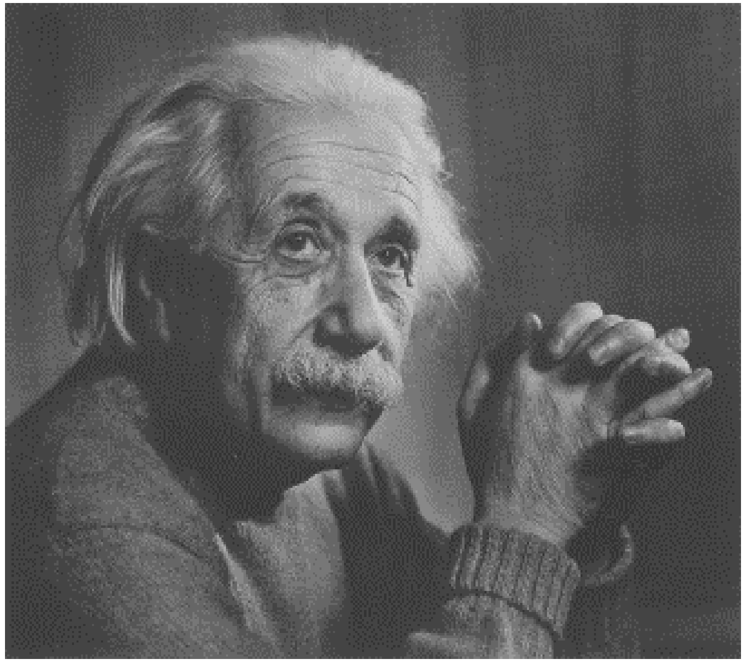
$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0, N-1) \\ f(1,0) & f(1,1) & \cdots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1, N-1) \end{bmatrix}.$$

Representing Digital Images contd..

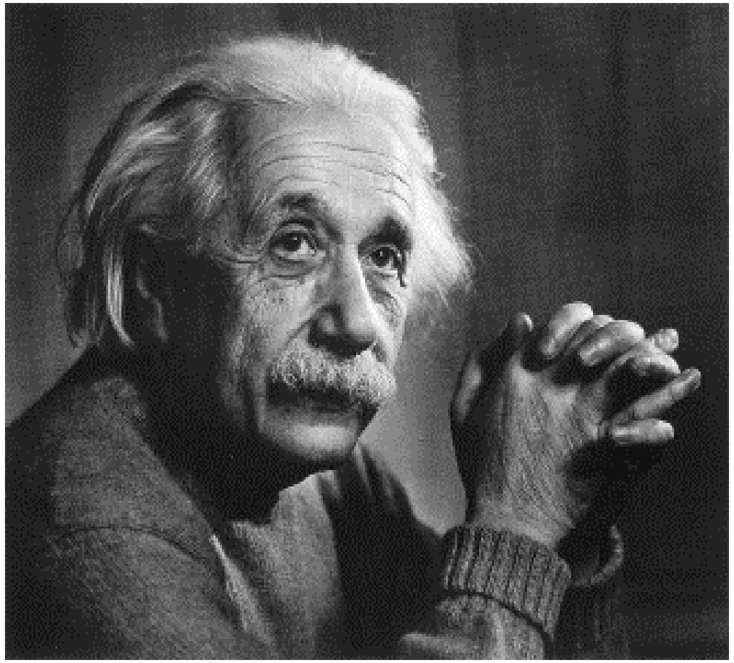
- Digitization Image: M rows, N columns, and, L discrete gray levels
  - *M* and *N*, have to be positive integers
- Processing, storage, and sampling hardware considerations: $L = 2^k$
- The range [0, L-1] → dynamic range of an image
- Images whose gray levels span a significant portion of the gray scale are referred to as having a high dynamic range
  - An image with high dynamic range tends to have high contrast
  - An image with low dynamic range tends to have low contrast (a dull, washed out gray look)



Low contrast



Medium contrast



High contrast

Representing Digital Images contd..

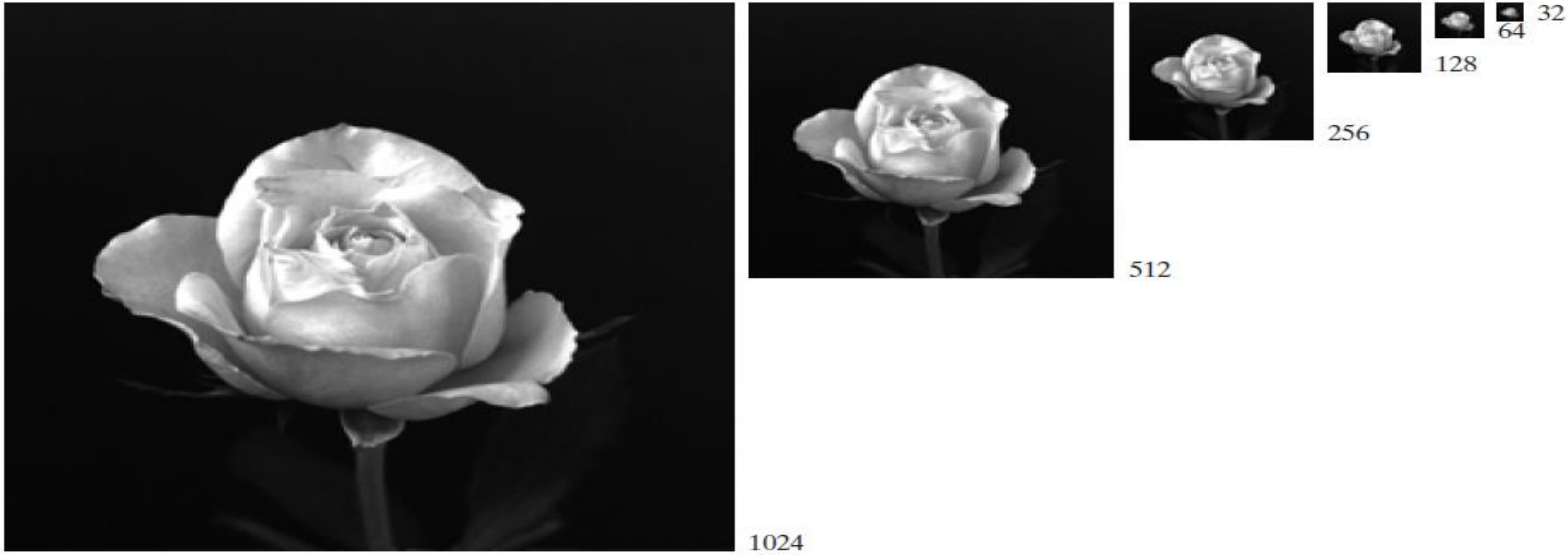
- The number, b, of bits required to store a digitized image is
  - b = M X N X k
- When M=N, b = N^2 X k

Number of storage bits for various values of N and k.

N/k	1 (L = 2)	2 (L = 4)	3 (L = 8)	4 (L = 16)	5 (L = 32)	6 (L = 64)	7 (L = 128)	8 (L = 256)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

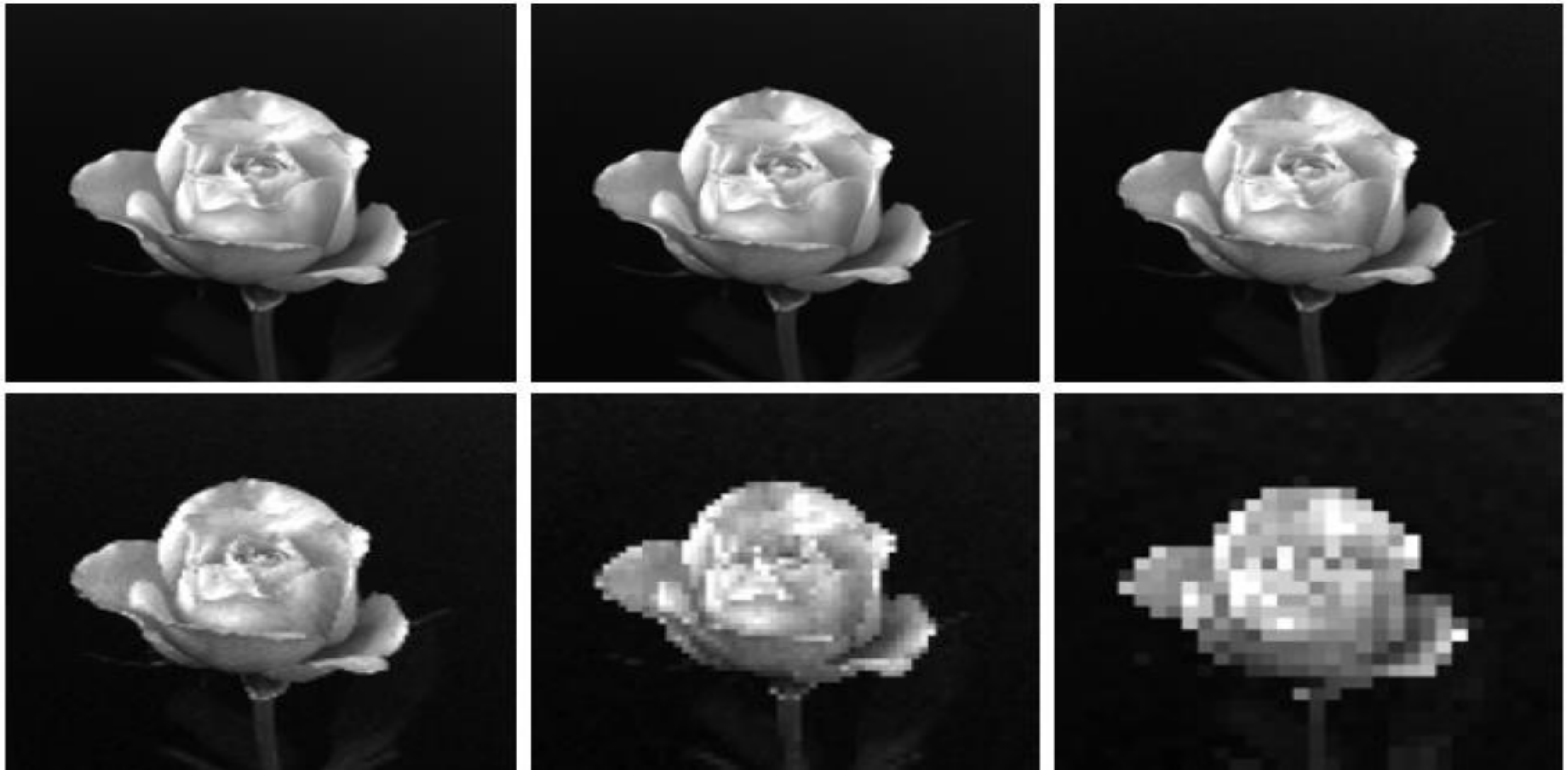
Spatial Resolution

- Sampling is the principal factor determining the spatial resolution of an image
- Spatial resolution is the smallest discernible detail in an image
- Dots per Inch (dpi) is a measure of Image resolution



Spatial Resolution contd..

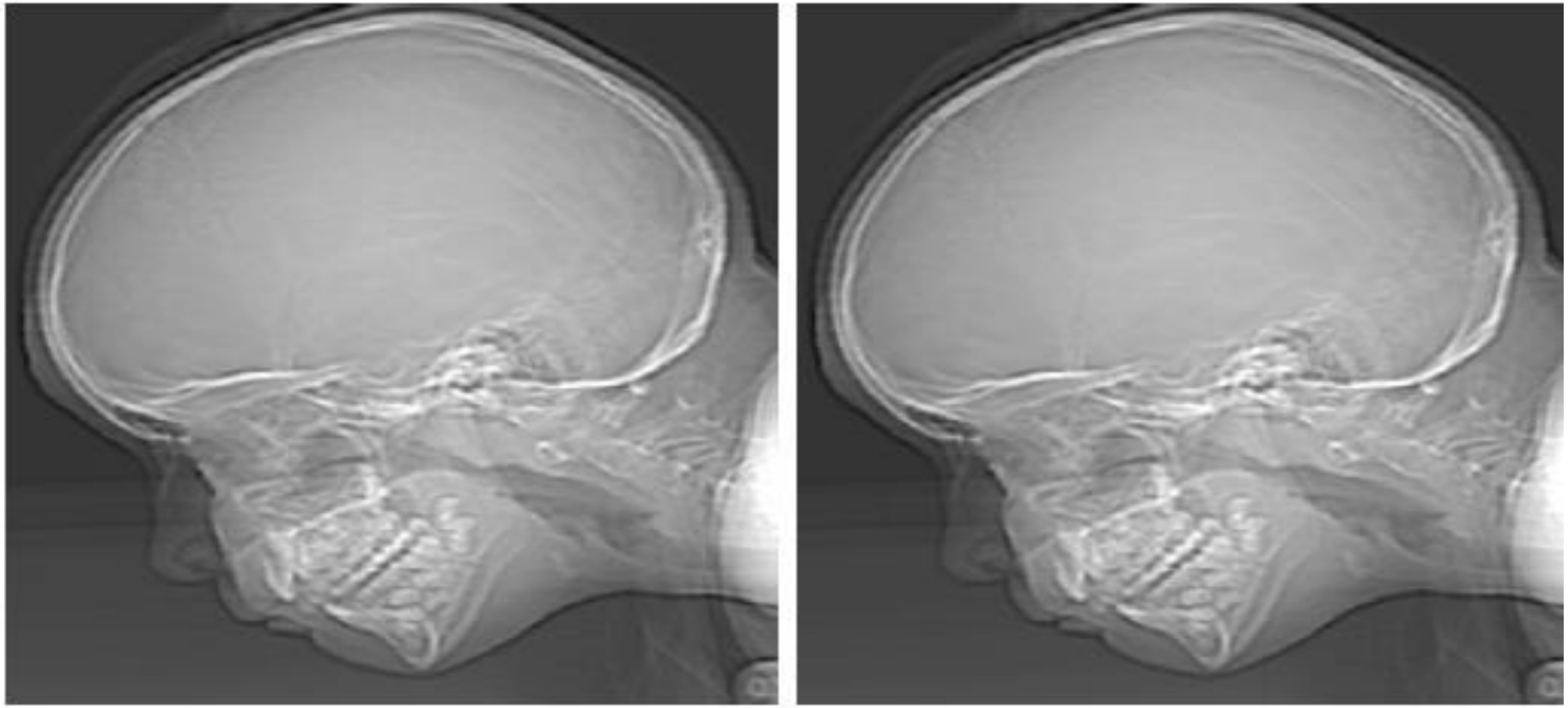
- The simplest way to compare the effects is to bring all the subsampled images up to size 1024*1024 by row and column pixel replication
- Checker board effect – (f)



(a) 1024*1024, 8-bit image.
(b) 512*512 image resampled into 1024*1024 pixels by row and column duplication.
(c) through (f) 256*256, 128*128, 64*64, and 32*32 images resampled into 1024*1024 pixels

Gray Level Resolution

- Keep the number of samples constant and reduce the number of gray levels from 256 to 2, in integer powers of 2

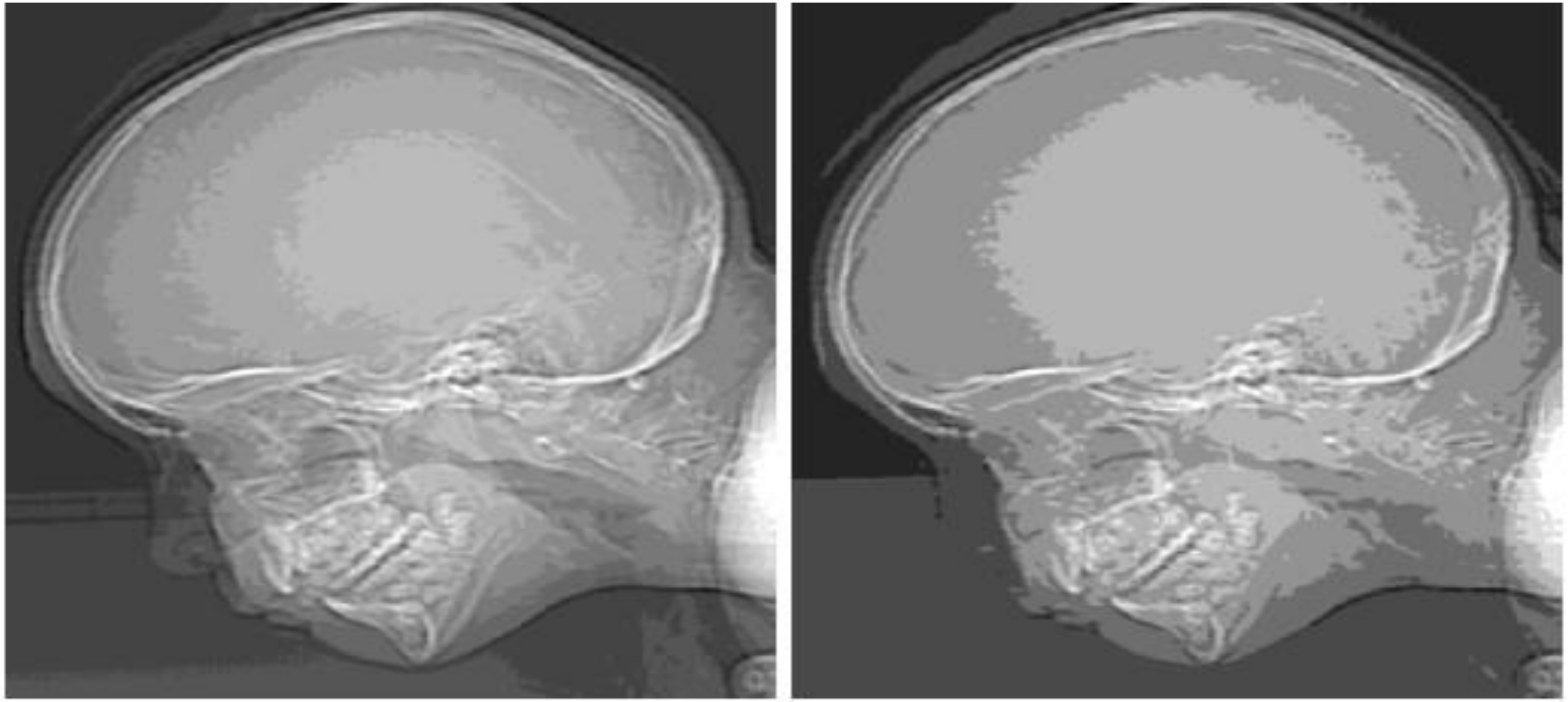


(a) 452*374, 256-level image. (b) Image displayed in 128 gray levels, while keeping the spatial resolution constant.

Gray Level Resolution contd..

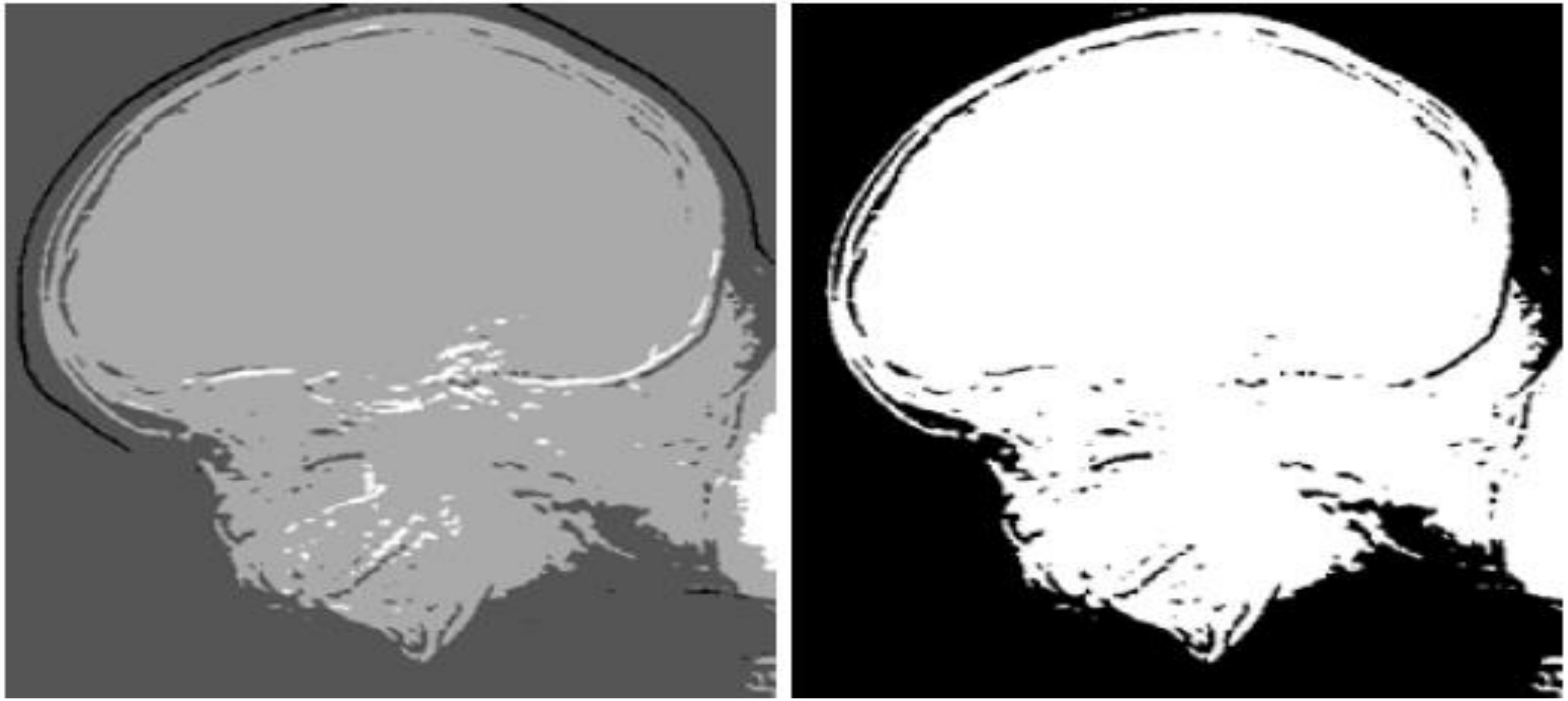
(a) Image with 64 gray levels
(b) Image displayed in 32 gray levels, while keeping the spatial resolution constant.

Gray Level Resolution contd..



(a) Image with 16 gray levels
(b) Image displayed in 4 gray levels, while keeping the spatial resolution constant.

Gray Level Resolution contd..



(a) Image with 4
gray levels
(b) Image
displayed in 2
gray
levels, while
keeping the
spatial
resolution
constant.

Gray Level Resolution contd..

- The 256-, 128-, and 64-level images are visually identical
- The 32-level image: Has an almost imperceptible set of very fine ridge-like structures in areas of smooth gray levels
  - This effect is caused by the use of an insufficient number of gray levels in smooth areas of a digital image
  - It is false contouring effect (As the ridges resemble topographic contours in a map)

Relationship between N and k

A rough rule of thumb : 256x256 pixels with 64 gray levels

Subjective study by Huang

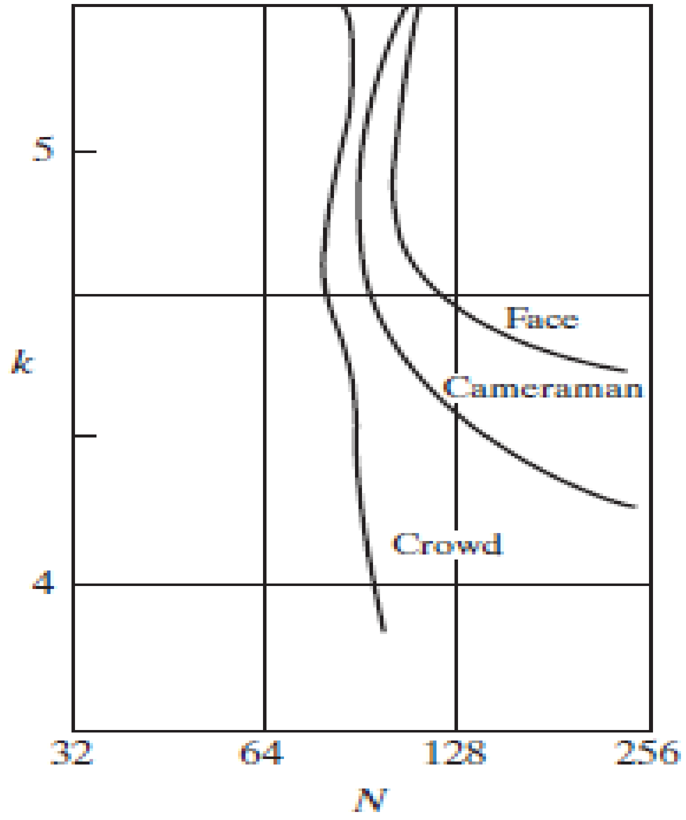
(a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail.

Relationship between N and k

- Sets of these three types of images were generated by varying *N* and k
- Observers were asked to rank them according to their subjective quality
- Results were summarized in the form of so-called *isopreference curves* in the Nk-plane

•**Curves tended to shift right and upward, but their shapes in each of the three image categories are similar**

•**Shift up and right in the curves implies for larger values of *N* and k, better will be the picture quality**



Zooming and Shrinking Digital Images

- Zooming may be viewed as oversampling, while shrinking may be viewed as undersampling

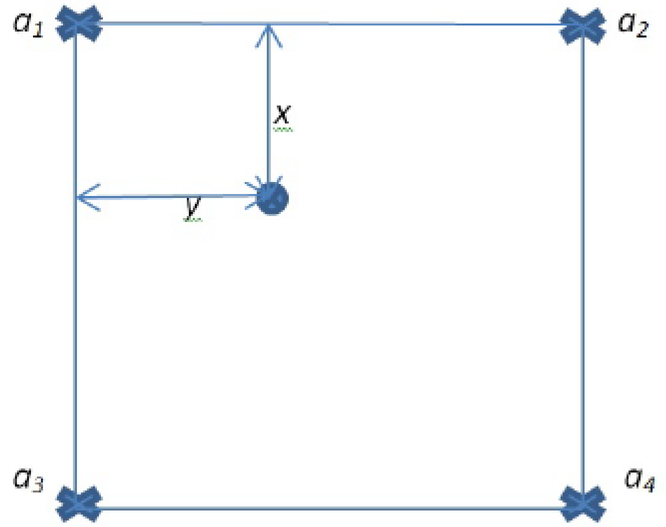
- Zooming requires two steps: the creation of new pixel locations, and the assignment of gray levels to those new locations

- Gray level assignment can be done using:
  - Nearest neighbor interpolation
  - Bilinear interpolation
  - Bicubic interpolation

Zooming and Shrinking Digital Images contd..

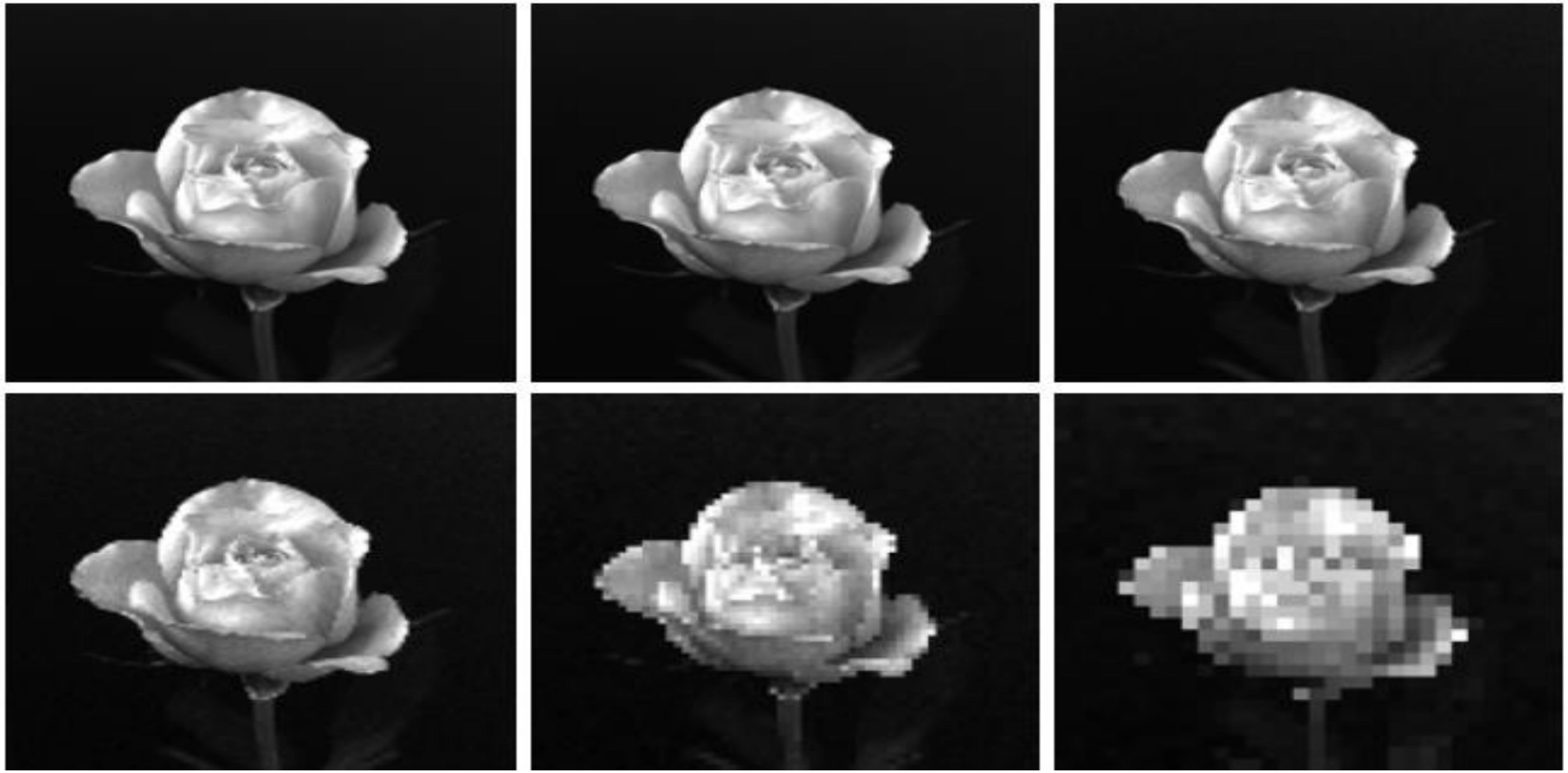
- Nearest neighbor interpolation is fast, but it produces a checkerboard effect
- Bilinear interpolation uses four nearest neighbors of a point
  - ➢ Sophisticated way of accomplishing gray-level assignments

$$v(x, y) = ax + by + cxy + d$$



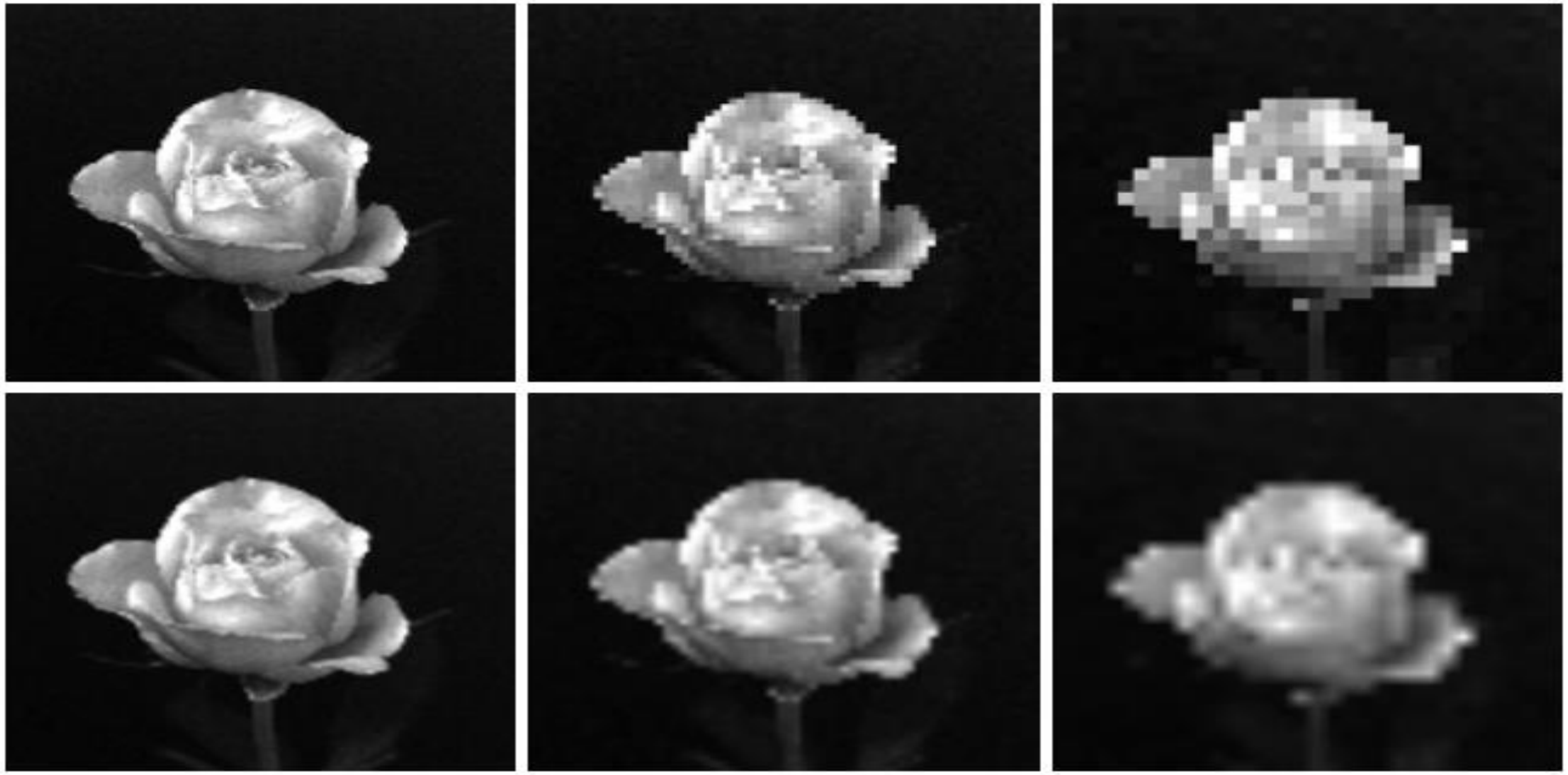
Zooming and Shrinking Digital Images contd..

- Pixel replication – duplicate each column and row to enlarge the image



Top row: images zoomed from 128X128, 64X64, and 32X32 pixels to 1024X1024 pixels, using nearest neighbor gray-level interpolation

Bottom row: same sequence, but using bilinear interpolation.



Basic Relationships Between Pixels

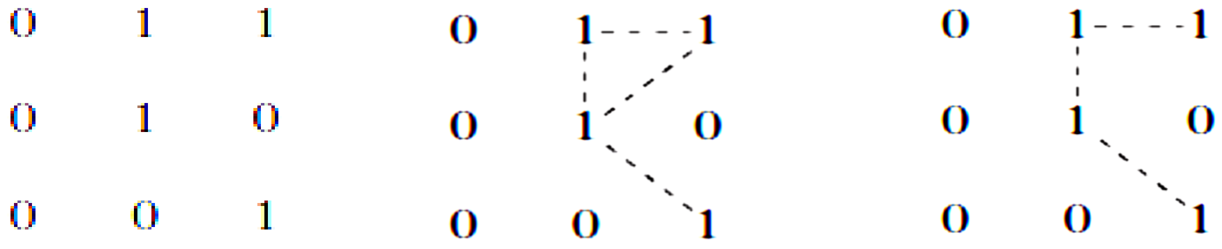
- **Neighbors of a Pixel**
  - A pixel p at coordinates (x, y) has four neighbors (two *horizontal* and two *vertical* neighbors)
  - $N_4(p)$: 4-*neighbors* of p - (x+1, y), (x-1, y), (x, y+1), (x, y-1)
  - Each pixel is at unit distance from (x, y)
  - $N_D(p)$: 4-*diagonal* neighbors of p - (x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)
  - $N_8(p)$: 8 neighbors $N_4(p) \cup N_D(p)$

Basic Relationships Between Pixels contd..

- **Adjacency**
  - 4-*adjacency:* Two pixels p and q with values from *V* are 4-adjacent if q is in the set N_4(p)
  - 8-*adjacency:* Two pixels p and q with values from *V* are 8-adjacent if q is in the set N_8(p)
  - *m-adjacency* (mixed adjacency): Two pixels p and q with values from *V* are m-adjacent if
    - q is in N_4(p), or
    - q is in N_D(p) and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from *V*

Basic Relationships Between Pixels contd..

- Mixed adjacency is a modification of 8-adjacency
  - Introduced to eliminate the ambiguities that often arise when 8-adjacency is used
- e.g.,:



(a) (b) (c)

Ambiguity in (b) is eliminated in (c)

Basic Relationships Between Pixels contd..

- Two image subsets S_1 and S_2 are adjacent (4 or 8 or m) if some pixel in S_1 is adjacent to some pixel in S_2

					S_1					S_2				
0	0	0	1	1	1	0	0	1	1	1	1	1	1	0
0	0	0	1	1	1	1	0	1	1	1	1	1	1	0
0	0	0	1	1	1	0	1	1	1	1	1	1	1	0
0	0	0	1	1	1	0	1	1	1	1	1	1	1	0
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0

Distance Measures

- For pixels p, q, and z, with coordinates (x, y), (s, t), and (v, w), respectively, *D* is a *distance function* or *metric* if
  - **(a)** $D(p, q) \geq 0$ ($D(p, q)=0$ iff $p=q$)
  - **(b)** $D(p, q)=D(q, p)$, and
  - **(c)** $D(p, z) \leq D(p, q)+D(q, z)$

- The *Euclidean distance* between p and q is defined as

$$D_e(p, q) = \left[(x - s)^2 + (y - t)^2\right]^{\frac{1}{2}}.$$

- For Euclidean distance measure, the pixels having a distance less than or equal to some value *r* from (x, y) are the points contained in a disk of radius r centered at (x, y)

- The D_4 *distance* (also called *city-block distance*) between p and q is defined as
    - $D_4(p, q) = |x - s| + |y - t|$
- The pixels having a D_4 distance from (x, y) less than or equal to some value r form a diamond centered at (x, y)
- The pixels with D_4 distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance:

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

The pixels with D_4=1 are the 4-neighbors of (x, y)

- The D_8 *distance* (also called *chessboard distance*) between p and q is defined as
  - $D_8(p, q) = \max(|x - s|, |y - t|)$

- In this case, the pixels with D_8 distance from (x, y) less than or equal to some value r form a square centered at (x, y)

- The pixels with D_8 distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance

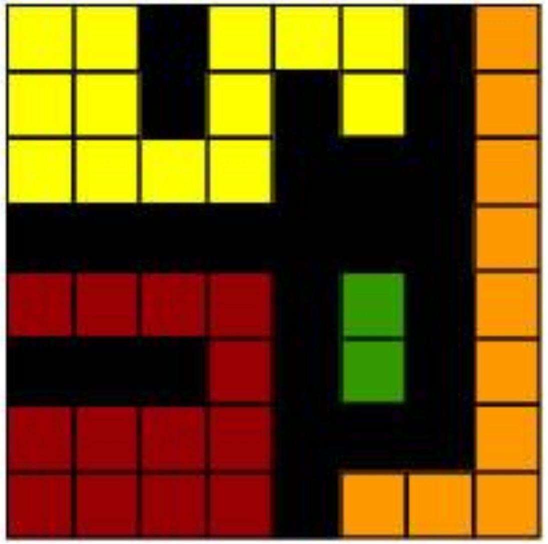
The pixels with D_8=1 are the 8-neighbors of (x, y)

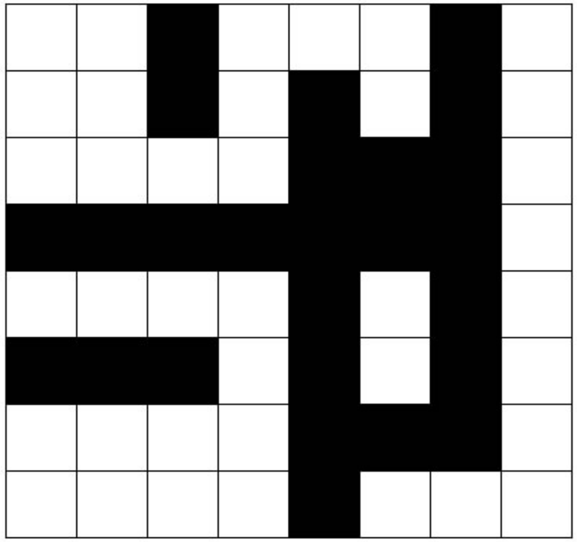
2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Connected Component Labeling: 4-connectivity and V = 1

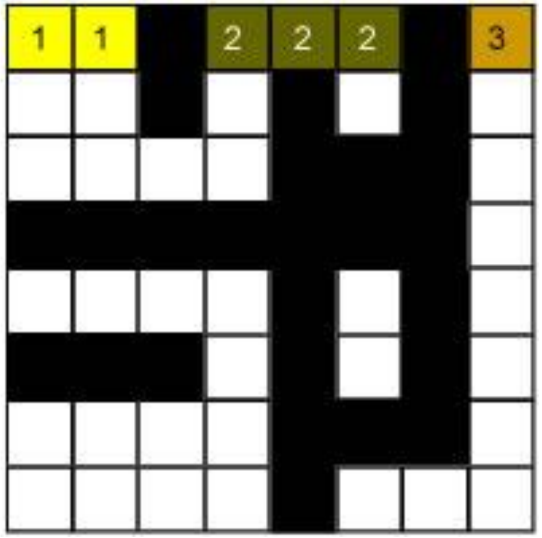
- Imagine scanning an image pixel by pixel, from left to right, and top to bottom
- Let p denote the pixel at any step in scanning process, and r and t, the upper and left hand neighbors of p
- If p = 0, move to the next scanning position
- If p = 1, examine r and t
  - If both are 0, assign a new label
  - If one of them is 1, assign its label to p
  - If both are 1 and have the same labels, assign it to p
  - If both are 1 and have different labels, assign one of the labels to p and make note that the two are equivalent
- Do a second pass by considering the equivalent classes

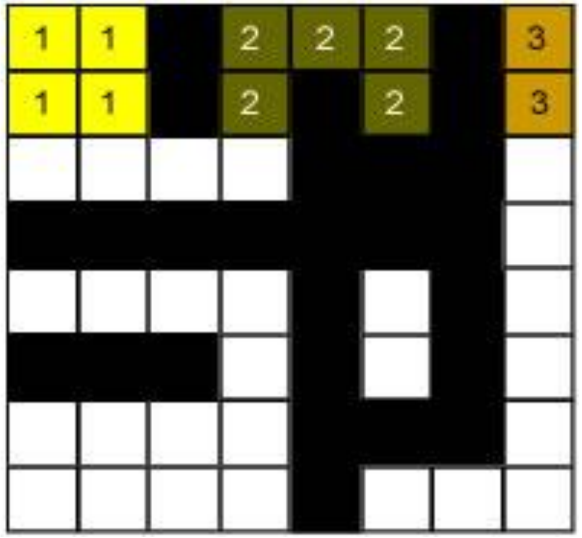
Connected Component Labeling: 4-connectivity and V = 1

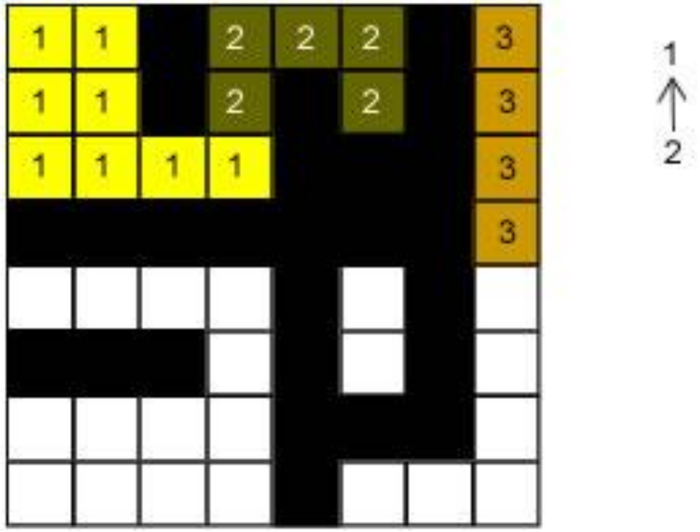




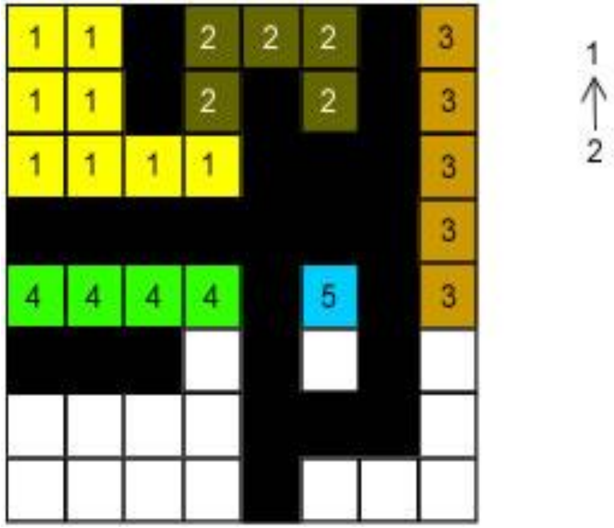
1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	1	1	0	1	1	1

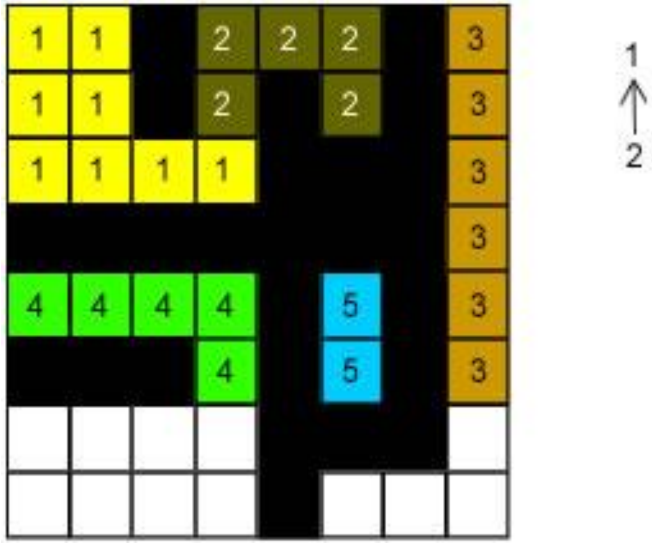


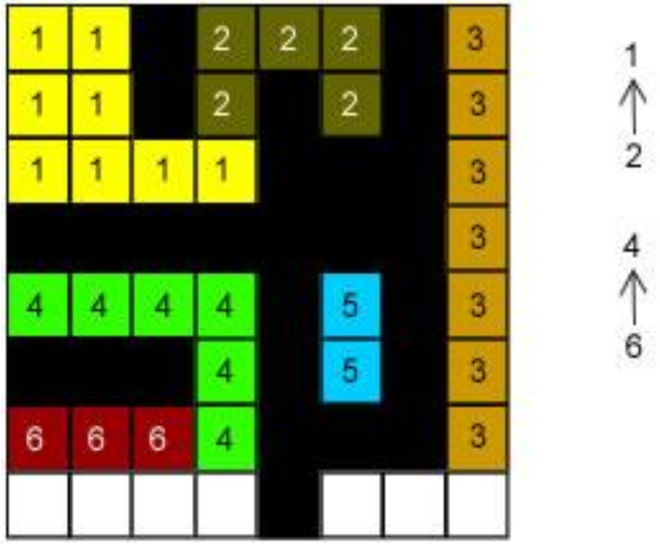


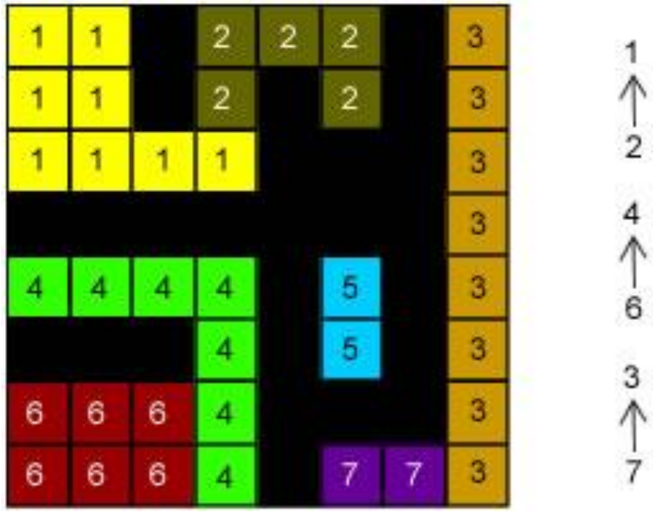


Connected Component Labeling: 4-connectivity and V = 1



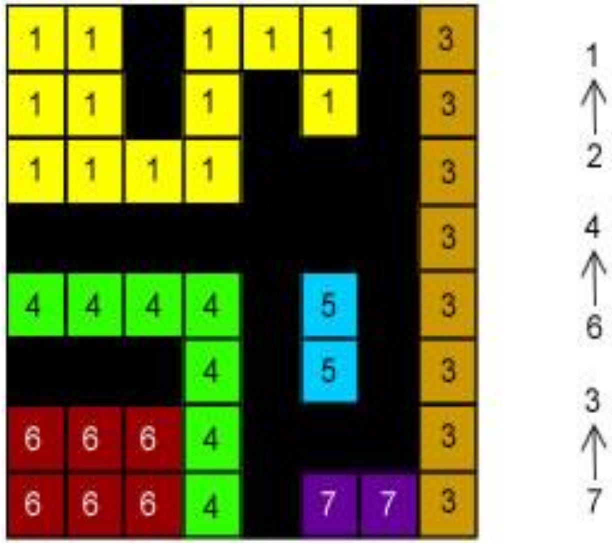


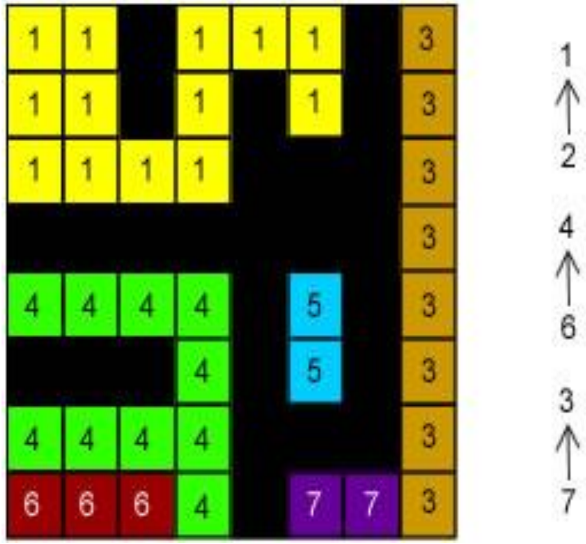


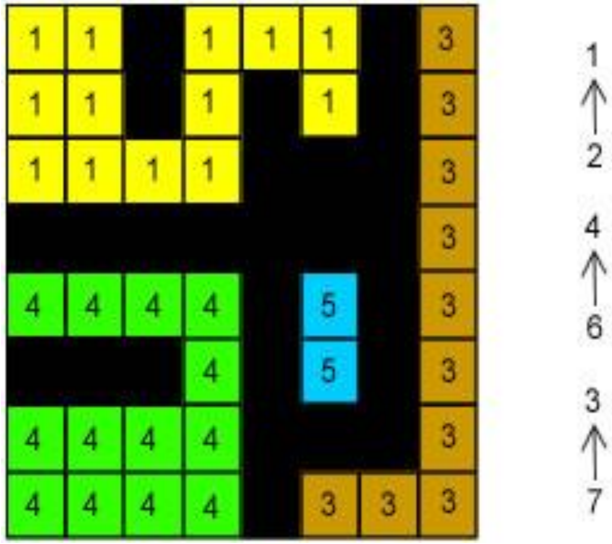


Connected Component Labeling: 4-connectivity and V = 1

- Second Pass







Connected Component Labeling: 8-connectivity and V = 1

- Let p denote the pixel at any step in scanning process, r and t, the upper and left hand neighbors, q and s, the upper diagonal neighbors of p
- If p = 0, move to the next scanning position
- If p = 1, examine q, r , s, and t,
  - If all are 0, assign a new label
  - If only one of them is 1, assign its label to p
  - If two or more neighbors are 1, assign one of the labels to p and make note of the appropriate equivalent classes
- Do a second pass by considering the equivalent classes

Image Processing Operations

- H is said to be a linear operator if, for any two images f and g and any two scalars a and b,

$$H(af + bg) = aH(f) + bH(g)$$

- The result of applying a linear operator to the sum of two images is identical to
  - Applying the operator to the images individually,
  - Multiplying the results by the appropriate constants, and then adding those results

An operator that fails the test of above equation is by definition *nonlinear*

- Arithmetic operations :
  - Addition $s(x, y) = f(x, y) + g(x, y)$
  - Subtraction $s(x, y) = f(x, y) - g(x, y)$
  - Multiplication $s(x, y) = f(x, y) X g(x, y)$
  - Division $s(x, y) = f(x, y) \div g(x, y)$

- Image Addition (Averaging) for Image Denoising

  - A noisy image is given by

$$g(x, y) = f(x, y) + \eta(x, y)$$

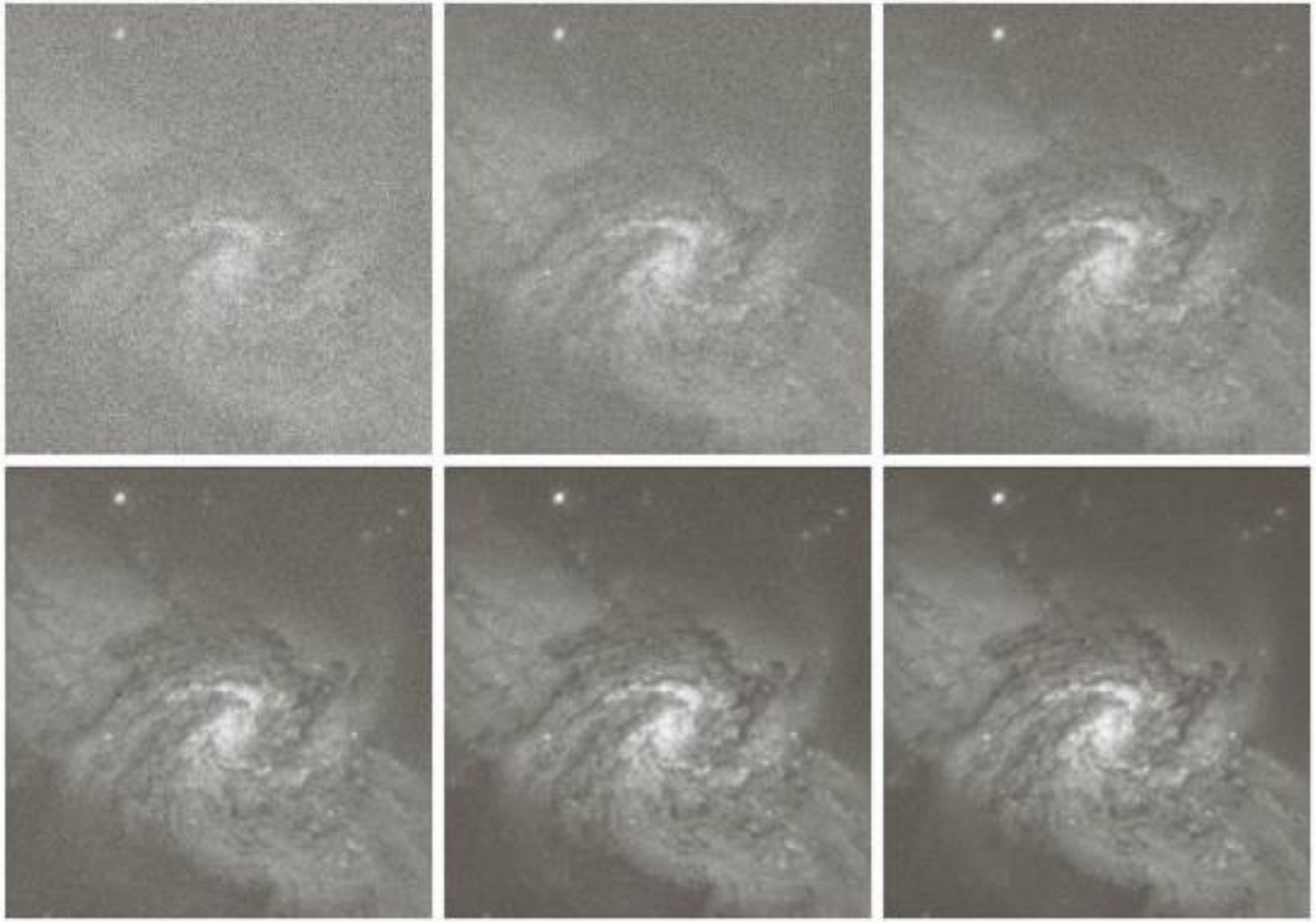
where $g(x, y)$: Corrupted image, $f(x, y)$: noiseless image, and noise $\eta(x, y)$ is assumed to be uncorrelated and zero mean

  - Averaging K noisy images

$$\overline{g}(x, y) = \frac{1}{K}\sum_{i=1}^{K} g_i(x, y)$$

then

$$E\{\overline{g}(x, y)\} = f(x, y) \qquad \sigma^2_{\overline{g}(x,y)} = \frac{1}{K}\sigma^2_{\eta(x,y)}$$



a b c
d e f

FIGURE 2.26 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)–(f) Results of averaging 5, 10, 20, 50, and 100 noisy images, respectively. (Original image courtesy of NASA.)

- Example of Image Subtraction

FIGURE 2.27 (a) Infrared image of the Washington, D.C. area. (b) Image obtained by setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity.

- Example of Image Subtraction

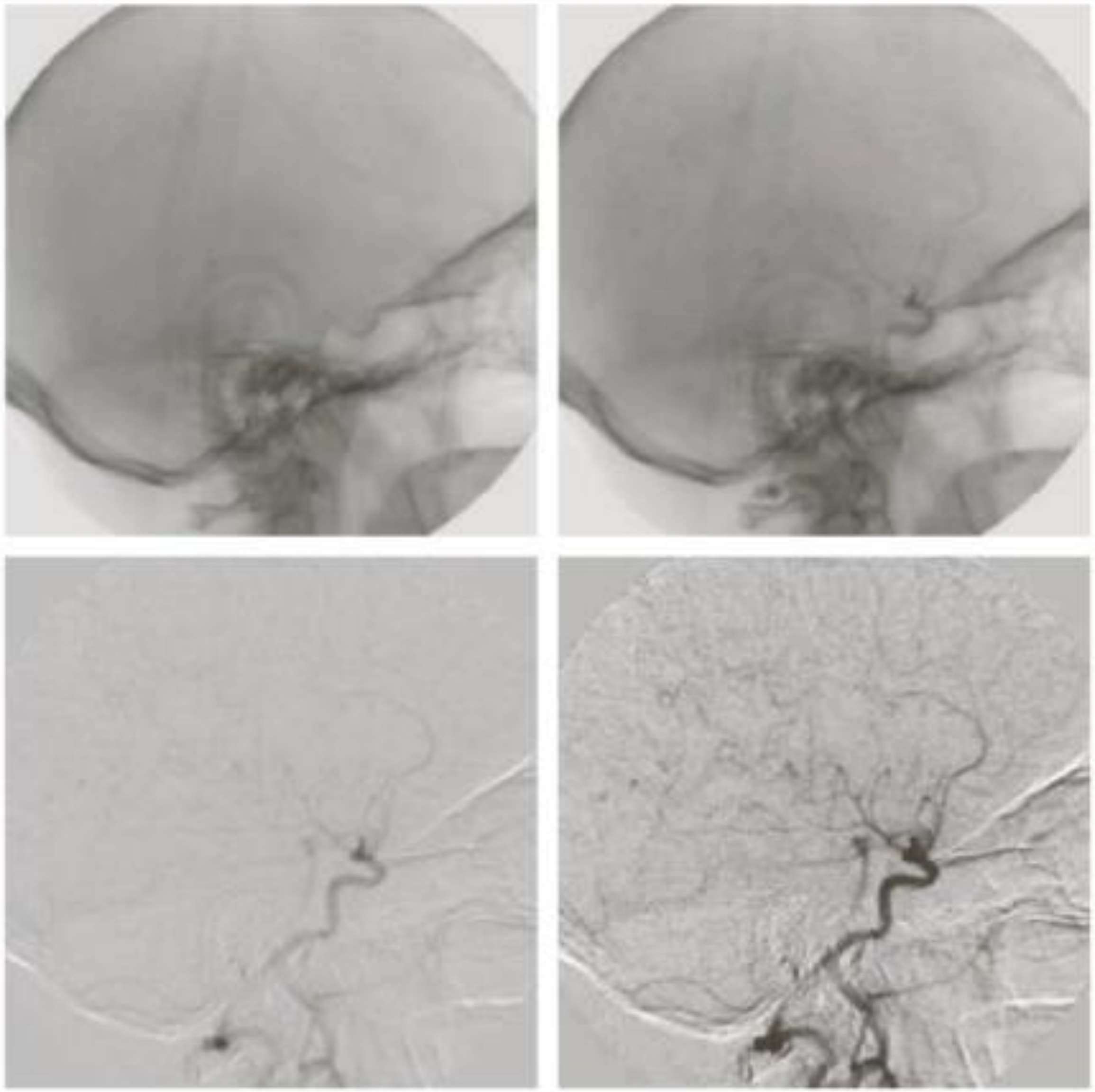
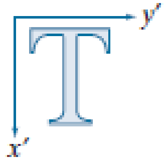
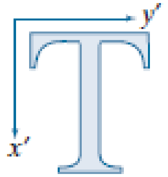
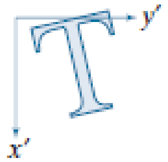
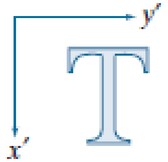
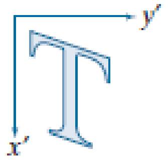


FIGURE 2.28 Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of The Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)

Geometric Spatial Transformations

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to −1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x\cos\theta - y\sin\theta$ $y' = x\sin\theta + y\cos\theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$